



NEVON SOLUTIONS
Information Technology

Software Requirements Specification

Biometric Student Attendance System

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Declaration

Declaration of Use of AI Tools

Nevon Solutions maintains internal guidelines governing the responsible and ethical use of Artificial Intelligence (AI) tools in the preparation of project documentation and professional deliverables.

In the development of this Software Requirements Specification (SRS) for the Biometric Student Attendance System, AI-assisted tools (e.g., ChatGPT) were utilised solely as supplementary support mechanisms to improve linguistic clarity, document structure, formatting consistency, and editorial refinement. These tools were used to enhance readability and assist with drafting where appropriate.

All requirements elicitation, analytical assessments, requirements modelling, stakeholder interpretation, validation decisions, technical evaluations, and final requirement definitions contained within this SRS were independently performed, reviewed, and approved by the Nevon Solutions Business Analysis Department. No requirements, system specifications, or analytical conclusions were autonomously generated by AI systems.

Nevon Solutions assumes full accountability for the accuracy, completeness, authenticity, and professional integrity of the contents of this SRS. The use of AI tools was conducted in compliance with applicable academic integrity policies, professional standards, and organisational governance guidelines.

Declaration on Plagiarism and Originality

Nevon Solutions upholds stringent standards of originality, transparency, and professional ethics in all project deliverables.

This Software Requirements Specification (SRS) for the Biometric Student Attendance System has been independently developed by the Nevon Solutions Business Analysis Department in accordance with the approved Project Specification. The requirements analysis, stakeholder interpretations, functional and non-functional requirement definitions, modelling artefacts, traceability mappings, and supporting documentation presented herein represent the team's own professional analysis and judgement, except where external references have been explicitly cited and acknowledged.

This declaration is formally endorsed on behalf of the analysis team by the Project Lead, who has reviewed, validated, and approved the final version of this SRS document.

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Name (Project Lead)

Signature

03/03/2026

Date



1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) is produced through the Specifying method of the Requirements Engineering (RE) process, and defines the complete set of regulated-revision requirements for the Biometric Student Attendance System to be developed by Nevon Solutions for BioSec Educational Institution. This document covers Version 0.8 of the system and constitutes the primary contractual basis for design, development, testing, and delivery of the software product.

This SRS incorporates all business operations and technical requirements produced from the Analysing method, including breakdown, clarification, interpretation, and categorisation of the BioSec Project Specification (PS), together with Nevon Solutions' development provisions covering operating environment, quality model, requirements traceability, and change mitigation strategy.

1.2 Document Conventions

This SRS is based on the IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998) template by Karl E. Wiegers. Structural modifications have been made to better accommodate the project's requirements engineering methodology and contractual context. Specifically, the template has been extended to include a dedicated Inceptive Requirements Traceability Matrix (iRTM) section ([Appendix D](#)) and a Change Mitigation Strategies section ([Section 8](#)) to ensure traceability from PS to SRS and provide pre-approved mechanisms for managing requirements change after development commences.

The following conventions apply throughout this document:

- Requirement IDs prefixed BOR indicate Business Operations Requirements. Those prefixed TR indicate Technical Requirements.
- Level 0 requirements (BOR-0-xx, TR-0-xx) define platform-level constraints applicable across the entire system.
- Level 1 requirements define specific functional or technical behaviours within subsystem categories.
- Atomic requirements derived from compound decomposition carry a parent ID reference for traceability.
- The notation (Revised) against a requirement ID indicates the original PS clause was modified during Analysing. The notation (Retained) indicates it was accepted without change.
- Priority ratings are: High (must have for system launch), Medium (required before warranty expiry).

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following audiences:

- **Development team:** Read [Section 3](#), [Section 4](#), and [Section 5](#) for all functional and non-functional requirements. [Section 2.4](#) covers the operating environment and platform.
- **Project Manager:** Read [Section 2](#) for product description and [Section 6](#) for the iRTM and change mitigation strategy.
- **Quality Assurance team:** Read all sections. Pay particular attention to acceptance criteria within each requirement clause and the quality model in Section 5.4.
- **BioSec stakeholders:** Read [Section 1.4](#), [Section 2](#), and the [Section 9](#) for overall scope and approval.
- **Testers:** Read [Section 4](#) and [Section 5](#) for all testable requirements, and [Appendix B](#) for open issues to be re-evaluated during testing.



- **QC Auditors (Independent):** Read all sections of this SRS for detection of authorship errors, omissions of topics, and latent contradictions. Pay particular attention to consistency between requirements in Sections 3, 4, and 5, the Data Dictionary in [Appendix C](#), and traceability mappings in [Appendix D](#).
- **QA Auditors (Independent):** Read all sections, with focus on evidence of compliance with the Analysing and Specifying methods used throughout the RE process. Supporting process documentation, including the Analysing Report (NS-AR-2026-001), meeting minutes, and repository records, will be made available separately.

1.4 Product Scope

The Biometric Student Attendance System is a cloud-hosted software product commissioned by BioSec Educational Institution to replace its current manual paper-based attendance process across five campuses in Singapore. The system eliminates buddy signing by using biometric fingerprint verification to record student attendance at building entrances and main corridors.

The product comprises three components: an Android and iOS mobile application for parents and teachers, an administrator web portal, and a cloud backend hosted on Microsoft Azure. The system integrates with BioSec's Medical Certificate System (MCS) for leave request validation, Firebase and APNS for push notifications, and the School Management Database for student enrolment data.

The system supports attendance recording, leave request submission and approval, compliance threshold monitoring, and automated notifications. All biometric data is handled in compliance with Singapore's Personal Data Protection Act (PDPA).

1.5 References

- BioSec Project Specification, Document ID: BS-PS-2026-001, Version 1.0
- Nevon Solutions Analysing Report, Document ID: NS-AR-2026-001, Version 1.0, 8 February 2026
- Nevon Solutions GitHub Repository:
<https://github.com/wenxuannx/Biometric-Student-Attendance-System>
- IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998
- Singapore Personal Data Protection Act (PDPA) 2012, as amended
- ISO/IEC 19794-2: Biometric Data Interchange Formats

2. Overall Description

2.1 Product Perspective

The Biometric Student Attendance Management System is a new, self-contained product commissioned as a replacement for BioSec's manual attendance process. It is not a follow-on release or extension of an existing system. The system operates as a client-server architecture with cloud hosting on Microsoft Azure Singapore region.

The system interfaces with the following external systems: BioSec's Medical Certificate System (MCS) for leave validation, Firebase and Apple Push Notification Service (APNS) for mobile push notifications, an SMTP email server for compliance warning delivery, and the School Management Database for student enrolment data. The revised Context Diagram (NS-AR-2026-001, [Figure 1](#)) shows all external interfaces. Refer to Section 3 for detailed interface specifications.

2.2 Product Functions

The system performs the following major functions:



- Biometric attendance recording: fingerprint-based student identity verification at campus entry points, with manual override on repeated scan failure.
- Attendance data access: role-scoped mobile access for parents (child only) and teachers (assigned classes only), with periodic synchronisation.
- Leave request management: parent submission with MCS medical certificate validation, administrator review, and approval enforcement.
- Compliance monitoring: automatic compliance warnings when student attendance falls below the configured threshold, configurable by administrators.
- Push notification delivery: attendance alerts and leave status updates delivered via Firebase, APNS, with SMS fallback.
- Administrator web portal: centralised management of attendance records, leave requests, compliance reports, and user accounts.

2.3 User Classes and Characteristics

Three user classes interact with the system:

- Administrators: School staff responsible for system management, attendance oversight, leave approval, and compliance configuration. High technical literacy expected. Access via web portal only. Primary users for reporting and system configuration functions.
- Teachers: Academic staff who view attendance records for their assigned classes. Moderate technical literacy. Access via mobile application. Read-only attendance access with no approval authority.
- Parents: Guardians of enrolled students who view their child's attendance and submit leave requests. Variable technical literacy; interface must be intuitive. Access via mobile application only.

Additionally, Students are indirect participants in the system. They are not system users and do not authenticate or access any interface. Students interact with the system solely through the biometric fingerprint scanners at campus entry points, where their scan data is captured to generate attendance records. Students are classified as data subjects under PDPA.

2.4 Operating Environment

The system shall operate in the following environment:

- Cloud platform: Microsoft Azure, Singapore region, with HTTPS and TLS 1.2 or higher for all endpoints.
- Mobile applications: Android (minimum Android 10) and iOS (minimum iOS 14), distributed via Google Play Store and Apple App Store respectively.
- Web portal: Accessible on current versions of Chrome, Safari, Firefox, and Edge. Desktop only, no mobile-responsive requirement.
- Database: Microsoft SQL Server, hosted on Azure.
- External integrations: BioSec MCS via HTTPS REST API with OAuth 2.0; Firebase and APNS for push notifications; SMTP for email delivery.

2.5 Design and Implementation Constraints

- All biometric data must be stored encrypted using AES-256 and transmitted using TLS 1.2 or higher.
- The system must comply with Singapore's PDPA for all biometric and personal data collection, storage, and processing.
- Mobile applications must be code-signed with BioSec institutional certificates prior to distribution.



- The MCS integration must use HTTPS REST with OAuth 2.0. No alternative authentication mechanism is permitted.
- Biometric template storage format is a pending open issue (see [Appendix B](#)). Development shall proceed with ISO/IEC 19794-2 until the open issue is resolved, with a kill date of 15 March 2026.
- The system must support offline attendance viewing and leave request drafting for up to 48 hours, with a 50MB local storage limit.

2.6 User Documentation

The following documentation shall be delivered with the system as per TR-1-DOC-01 to TR-1-DOC-04:

- Administrator User Manual: covering all web portal functions including user management, attendance records, leave approval, compliance configuration, and reporting.
- Teacher User Manual: covering mobile application functions for attendance viewing.
- Parent User Manual: covering mobile application functions for attendance viewing and leave request submission.
- API Documentation: covering all endpoints, request and response formats, authentication, and error codes.
- Database Documentation: covering ERD, schema definitions, and stored procedure documentation.
- Source Code Documentation: covering inline comments, README, and build guide.

All documentation shall be delivered in PDF format. Institutional ownership transfers to BioSec upon project completion.

2.7 Assumptions and Dependencies

- BioSec's MCS is assumed to be operational and its API documentation will be made available to Nevon Solutions before development of the MCS integration commences. Specific encryption handshake protocol for MCS remains an open issue (Refer to [Appendix B](#)).
- BioSec will provide institutional certificates for mobile application code signing before the distribution deadline.
- Student enrolment data in the School Management Database is assumed to be current and accurately structured. Data mapping specifications are a pending open issue.
- BioSec will confirm the acceptable Recovery Point Objective (RPO) and Recovery Time Objective (RTO) for attendance data by 20 February 2026. Disaster recovery design is contingent on this.
- All five campuses are assumed to have stable network connectivity to support biometric scan data transmission. Offline mode (48-hour limit) is the contingency for temporary outages.

3. External Interface Requirements

3.1 User Interfaces

The Biometric Student Attendance System provides the mobile application that serves parents and teachers, while the administrator web portal serves administrative staff. Access to each interface is restricted by role, enforced at both the application and API layers. (Refer to [Figure 2](#)).

General Interface Requirements

- **UR-GUI-01:** The administrator web portal shall be accessible on current versions of Chrome, Safari, Firefox, and Edge on desktop only, with no mobile-responsive requirement.
- **UR-GUI-02:** The system shall implement role-based access control at the interface layer, where each user class sees only the functions and data relevant to their assigned role.



- **UR-GUI-03:** The parent and teacher mobile application shall be mobile user friendly with simple navigation.

Parent and Teacher Mobile Application Interface Requirements

- **UR-MOB-01:** The parent interface shall display monthly attendance records for the parent's registered child only, with attendance status shown as Attended, Absent, Late, or Excused.
- **UR-MOB-02:** The parent interface shall provide a leave request submission screen that allows date selection, reason entry, and optional medical certificate upload for MCS validation.
- **UR-MOB-03:** The teacher interface shall display attendance records for assigned classes only, in a read-only format with no approval or submission functionality.
- **UR-MOB-04:** The mobile application shall display push notifications for absence events and leave status updates.
- **UR-MOB-05:** Primary tasks, including attendance viewing and leave request submission, shall be reachable within three interactions from the home screen, to accommodate users with varying technical literacy.
- **UR-MOB-06:** The parent and teacher authentication shall include a 2-Factor Authentication screen process.

Administrator Web Portal Interface Requirements

- **UR-WEB-01:** The administrator web portal shall provide a centralised dashboard displaying attendance status for all students across all five campuses, filterable by campus, class, student, and date range.
- **UR-WEB-02:** The portal shall display a leave review screen listing all pending leave requests with student name, date range, reason, and MCS validation status. Administrators may approve or reject requests, with an optional comment field.
- **UR-WEB-03:** The portal shall display a compliance review screen to highlight students whose attendance falls below the configured compliance threshold
- **UR-WEB-04:** The compliance review screen shall include buttons for administrators to issue warning letters to parents
- **UR-WEB-05:** The portal shall display a reports screen consisting of different report categories such as attendance summary, compliance report, leave requests, biometric scans. Each set of report will display an export button for administrators to export to CSV/PDF format
- **UR-WEB-06:** The portal shall display a user management screen for administrators to add, edit, or delete users. The screen will display a list view of users indicating their personal account details and status.
- **UR-WEB-07:** The portal shall allow administrators to configure the attendance compliance threshold percentage via the settings page.
- **UR-WEB-08:** the portal shall display an activity log for administrators to view system notifications
- **UR-WEB-09:** the portal shall include a 2-Factor Authentication screen during the authentication process.
- **UR-WEB-10:** The portal shall include a bulk selection of students via checkboxes in the student attendance records to implement manual attendance override.

3.2 Hardware Interfaces

The Biometric Student Attendance System interfaces directly with the biometric fingerprint scanning devices already procured and deployed by BioSec across all five campuses.

Hardware Technical Requirements



- **FR-HW-01 (from BOR-0-03):** The system shall receive biometric scan data from fingerprint devices at all five BioSec campuses, each identified by a unique device_id registered in the system prior to deployment.
- **FR-HW-02 (from TR-1-BE-04):** Scan processing time from physical fingerprint contact to attendance outcome display shall not exceed 2 seconds per attempt, supporting a throughput of 50 students per device per 10 minutes during peak periods.
- **FR-HW-03 (from TR-1-BE-01c):** The system shall support up to 200 concurrent biometric scans across all campus devices simultaneously without exceeding the 2-second processing threshold or degrading backend API response times.
- **FR-HW-04 (from BOR-1-ATT-01):** After 3 consecutive failed scan attempts, the system shall initiate a supervised manual override flow and generate a flagged attendance record for administrator review.
- **FR-HW-05 (from BOR-0-03):** If a biometric scanner device becomes unavailable or fails during a session, students who were not scanned shall not be automatically marked absent. The device failure shall be logged with the device_id and timestamp for administrator action.

3.3 Software Interfaces

The system establishes secure integration boundaries with external software components to support leave validation, notification delivery, email distribution, student enrolment data, and database operations.

Software Technical Requirements

- **FR-SW-01 (from TR-1-BE-02):** The system shall validate leave request medical certificates via the MCS REST API using HTTPS and OAuth 2.0 authentication before recording any leave submission.
- **FR-SW-02 (from TR-1-BE-02):** The system shall apply a 5-second timeout and 3 retry attempts with exponential backoff for all MCS API calls, logging failures to ExternalSystemLog and notifying the administrator on exhaustion.
- **FR-SW-03 (from TR-0-01):** The system shall deliver push notifications via Firebase Cloud Messaging for Android devices and APNS for iOS devices, covering attendance alerts and leave status updates.
- **FR-SW-04 (from BOR-1-WEB-01):** The system shall deliver compliance warning letters via an institutional SMTP email server to registered parent email addresses.
- **FR-SW-05 (from TR-1-BE-05):** The system database shall be hosted on Microsoft SQL Server on Azure SQL Database with AES-256 encryption for all biometric data, daily automated backups, and monthly restoration verification.

Institutional Database Integration

- **Microsoft SQL Server on Azure SQL Database:** Internal system database hosted on Azure SQL Database, Singapore region. Stores all attendance, leave, notification, and audit data across nine tables (Tables C.1 to C.9, Data Dictionary, see [Appendix C](#)). Accessed exclusively through the RESTful API layer. No direct external access permitted. [Encryption, backup, and restoration details governed by [FR-SW-05](#).]

External System Integration

- **Medical Certificate System (MCS):** HTTPS REST API with OAuth 2.0. Used for leave request medical certificate validation. Sends mc_reference and student_id; receives validation_status and mc_id. 5-second timeout, 3 retries, 99.5% SLA. Encryption handshake protocol is an open issue pending MCS vendor documentation.
- **Firebase Cloud Messaging (FCM):** FCM HTTP v1 API authenticated via service account credential. Sends notification payload to Android parent devices for absence alerts and leave status updates. Delivery status written to NotificationLog.
- **Apple Push Notification Service (APNS):** APNs HTTP/2 provider API authenticated via BioSec institutional APNs key. Sends notification payload to iOS parent devices. Delivery status written to NotificationLog. SMS gateway fallback triggered if delivery unconfirmed after 30 minutes.



- **SMTP Email Server:** SMTP over TLS to the BioSec institutional mail server. Sends compliance warning letters containing student attendance percentage and threshold details to registered parent email addresses. Server endpoint and configuration are open issues pending BioSec IT confirmation

Data Integrity and Auditability

- **Failure Isolation:** Integration failures with any external component shall not overwrite or invalidate existing AttendanceRecord or LeaveRequest entries.
- **Auditability:** All external system interactions are logged to ExternalSystemLog with system_name, request_timestamp, response_status, retry_attempt, and error_message for each event.

3.4 Communications Interfaces

Communication Functional Requirements

- **FR-COM-01 (from TR-1-CL-01c, BOR-1-MOB-02c):** Mobile applications shall synchronise attendance data with the backend every 5 minutes when connected, with a maximum data staleness of 30 minutes.
- **FR-COM-02 (from TR-0-01):** Push notifications shall be delivered via Firebase (Android) and APNS (iOS). If delivery is not confirmed within 30 minutes, the system shall fall back to SMS gateway delivery.
- **FR-COM-03 (from TR-1-BE-02):** MCS API communication shall comply with the interface contract defined in [FR-SW-01](#) and [FR-SW-02](#).
- **FR-COM-04 (from BOR-0-03):** All biometric scan data shall be transmitted from each scanner device to the cloud backend over the campus network using HTTPS.

4. System Features

4.1 Role-Based Access Control

4.1.1 Description and Priority

- The system will implement role-based access control (RBAC) for all three user classes (Administrators, Teachers, and Parents) on all five campuses. Each user class has access to only the functions and data required for their role. RBAC is a fundamental security control that must be implemented before any other system capability can be accessed, as mandated by PDPA and institutional access governance.
- Priority: High
- Reference: BOR-0-01, BOR-0-02, TR-1-CPR-01

4.1.2 Stimulus/Response Sequences

- After launching the application (mobile or web portal), the user will see a login screen. The user selects a role (parent, teacher, or administrator) and provides their credentials.
- The system authenticates users and restricts access based on their assigned role. Administrators are directed to a web portal that allows them complete control and configuration. Teachers are directed to the mobile application, which only allows them to view read-only attendance for assigned classes. Parents are directed to the mobile application, which has limited access to their child's attendance data and leave submission.
- If a user attempts to access a resource beyond their role scope, the system will deny the request and display an access denied message.
- If an administrator account is locked or deactivated, the system will prevent login and prompt the user to contact the institution.

4.1.3 Functional Requirements [UC-01, UC-02, UC-03]



- **FR-RBAC-01:** The system shall provide role-based access control for all three user classes: Administrators (web portal), Teachers (mobile, read-only attendance for assigned classes), and Parents (mobile, own child's data). Access restrictions will be enforced at both the application and API layers. (BOR-0-01, BOR-0-02, TR-1-CPR-01, UC-01, UC-02, UC-03)
- **FR-RBAC-02:** The system shall allow administrators to create new user accounts by assigning each account a unique ID, role (Administrator, Teacher, or Parent), and linked entity (student or class). (BOR-0-02, UC-01)
- **FR-RBAC-03:** The system shall allow administrators to update a user account's role assignment, reset credentials, and unlock a locked account. All changes will be recorded in the Activity Log. (BOR-0-02, TR-1-CPR-01, UC-02)
- **FR-RBAC-04:** The system shall allow administrators to deactivate user accounts. Deactivated accounts will be denied login access immediately and kept in the database for audit purposes. (BOR-0-02, BOR-0-03, UC-03)

4.2 User Authentication

4.2.1 Description and Priority

- The system shall authenticate parents and teachers before allowing access to any attendance-related feature. This feature is required for all mobile application capabilities and enforces PDPA-compliant access control.
- Priority: High.
- References: BOR-0-01, BOR-1-MOB-01, and TR-1-CPR-01.

4.2.2 Stimulus/Response Sequences

- After launching the mobile app, the user will see the login screen.
- The user enters their username and password, then submits.
- The system validates credentials against the School Management Database and generates a session token if successful.
- The system displays the appropriate home screen for each role (parent or teacher).
- If credentials are invalid, the system will display an error message and prompt re-entry. After five unsuccessful attempts, the account is locked and the user is notified.

4.2.3 Functional Requirements

- **FR-AUTH-01:** The system shall validate the users' credentials before enabling access to attendance functionality. (BOR-1-MOB-01)
- **FR-AUTH-02:** Upon successful authentication, the system shall generate a session token and terminate it after a configurable period of inactivity. (TR-1-CPR-01)

4.3 Attendance Information Access

4.3.1 Description and Priority

- Authenticated users will have access to attendance records based on their role. Parents can only view their own child's records, while teachers can only view records for their assigned classes. Accurate attendance visibility is a key deliverable for Goals 2 and 3.
- Priority: High.
- Reference: BOR-1-MOB-02a, BOR-1-MOB-02b, BOR-1-MOB-02c, BOR-0-01, TR-1-CL-01c, and TR-1-CL-01f

4.3.2 Stimulus/Response Sequences

- The authenticated parent or teacher accesses the attendance screen.
- The system retrieves the user's role and displays attendance records limited to the authorised scope based on the child.



- The system displays monthly records with attendance status (Excused, Attended, Late, Absent).

4.3.3 Functional Requirements [UC-08, UC-09, UC-13, UC-15]

- **FR-ATT-01:** The system shall allow parents to view attendance data for their registered child at the application and API layers. (BOR-1-MOB-02a; TR-1-CPR-01, UC-09)
- **FR-ATT-02:** The system shall allow teachers to view attendance data for their assigned classes, which is enforced at the application and API layers. (BOR-1-MOB-02b, UC-13)
- **FR-ATT-03:** Following a successful biometric scan or an approved manual override, the system shall create an AttendanceRecord entry for each student, storing the student ID, session ID, campus, timestamp, and capture method. (BOR-1-ATT-01, BOR-1-ATT-02, UC-15)
- **FR-ATT-04:** The system shall not allow the deletion of AttendanceRecord entries. Deletion attempts will be rejected and recorded. (BOR-0-03, BOR-1-ATT-02, UC-08)

4.4 Leave Submission

4.4.1 Description and Priority

- The system shall allow parents to submit leave requests for their child's attendance-related absences, with mandatory MCS medical certificate validation where applicable. This feature directly contributes to Goal 4 by automating MC validation and eliminating manual processing.
- Priority: high.
- Reference: BOR-1-MOB-03a, BOR-1-MOB-03b, and TR-1-BE-02

4.4.2 Stimulus/Response Sequences

- Authenticated parents can submit leave requests by selecting the date(s) and reason for absence.
- If a medical certificate is required, parents upload it.
- The parent submits a leave request.
- The system sends the MC to the MCS REST API for validation. The system waits up to 5 seconds for a response before retrying up to three times with exponential backoff if failure occurs.
- Upon successful MCS validation, the system records the leave request as Pending and alerts the administrator.

4.4.3 Functional Requirements [UC-10,UC-11]

- **FR-LEAVE-01:** The system shall allow authenticated parents to submit a leave request for their registered child. (BOR-1-MOB-03a, UC-10)
- **FR-LEAVE-02:** The system shall submit the medical certificate to the MCS for validation before recording a leave request. A leave request shall not be recorded if MCS validation fails or returns an unresolved status. (BOR-1-MOB-03b; TR-1-BE-02) [Protocol and retry details governed by [FR-SW-01](#) and [FR-SW-02](#).]
- **FR-LEAVE-03:** The system shall record all leave submissions with a timestamp, user ID, student ID, leave reason, and MCS validation outcome for audit purposes. (BOR-1-MOB-03a, TR-1-BE-03)
- **FR-LEAVE-04:** The system shall allow authenticated parents to view the status and details of all leave requests submitted for their child, including the MCS validation outcome and administrator decision. (BOR-1-MOB-03a, BOR-1-MOB-03b, UC-11)

4.5 Attendance Notification

4.5.1 Description and Priority



- The system will send automated push notifications to parents regarding attendance-related events and leave status updates. Real-time parental notification within 15 minutes is a primary project goal (Goal 3).
- Priority: high.
- Reference: BOR-1-MOB-04 and TR-0-01.

4.5.2 Stimulus/Response Sequences [UC-11, UC-13]

- The biometric attendance capture system marks a student as absent from a scheduled session.
- The system sends an absence notification to the parent's registered mobile device using Firebase (Android) or APNS (iOS).
- If push delivery fails after 30 minutes, the system switches to SMS gateway delivery.
- When an administrator approves or rejects a leave request, the system sends a push notification to the parent with the decision.
- Parents receive and view notifications on their device.

4.5.3 Functional Requirements [UC-12, UC-14, UC-16]

- **FR-NOTIF-01:** The system shall notify parents via push notification within 15 minutes of a student failing to record attendance for a scheduled session. (BOR-1-MOB-04, UC-12).
- **FR-NOTIF-02:** The system shall notify parents via push notification of any changes in their leave request status (approved or rejected). (BOR-1-MOB-04, UC-12)
- **FR-NOTIF-03:** For each notification dispatched, the system shall create a Notification record in the database, including the recipient user ID, notification type, content, dispatch timestamp, delivery channel, and status. (BOR-1-MOB-04, UC-12, UC-14)
- **FR-NOTIF-04:** The system shall allow parents to view their notification history, which will include all attendance and leave status notifications received, ordered by most recent. (BOR-1-MOB-04, BOR-1-MOB-02a, UC-12)
- **FR-NOTIF-05:** The system shall update the delivery status of a Notification record when it is confirmed by Firebase, APNS, or the SMS gateway, or when the fallback is activated. (BOR-1-MOB-04, UC-12, UC-14)
- **FR-NOTIF-06:** Notification records must be retained for at least 12 months before being archived or purged in accordance with the institution's data retention policy. (BOR-0-03, BOR-1-MOB-04)
- **FR-NOTIF-07:** The system shall notify the administrator via the web portal when a new leave request is submitted by a parent and is pending review, displaying the student name, submission timestamp, and MCS validation status. (BOR-1-WEB-02a, BOR-1-MOB-03a, UC-16)
- **FR-NOTIF-08:** The system shall notify the administrator via the web portal when a biometric scanner device fails during a session, identifying the device_id, campus location, and failure timestamp, so that administrator action can be taken. (BOR-0-03, FR-HW-05, UC-16)
- **FR-NOTIF-09:** The system shall notify teachers via push notification of attendance events for students in their assigned classes, including absence and late arrival alerts. (BOR-1-MOB-04, UR-MOB-04, UC-14)

4.6 Attendance Oversight and Compliance Monitoring

4.6.1 Description and Priority

- The system will provide administrators with a centralised web portal view of attendance and compliance information across all five campuses. When students' attendance falls below the configured threshold, automated compliance warnings will be issued. Centralised oversight is required to manage multi-campus attendance integrity, which supports Goal 5.
- Priority: High
- References: BOR-1-WEB-01a, BOR-1-WEB-01b, TR-0-02, TR-1-BE-01b.



4.6.2 Stimulus/Response Sequences

- The administrator logs in to the web portal and views the attendance dashboard.
- The system displays a summary of attendance status for selected campuses, classes, and dates.
- The administrator uses filters (campus, class, date, student) to view detailed attendance records.
- Identifies and highlights students whose attendance percentage falls below the configured compliance threshold.
- The administrator sets or updates the compliance threshold.
- Compliance warning letters are automatically generated and emailed to registered parent addresses.
- The administrator generates an attendance report, which the system returns within 10-60 seconds depending on data volume.

4.6.3 Functional Requirements [UC-04, UC-05, UC-06, UC-08, UC-17]

- **FR-OVS-01:** The system shall provide administrators with a web portal that displays attendance status for all students across five campuses, filterable by campus, class, student, and date range. (BOR-1-WEB-01a, UC-08)
- **FR-OVS-02:** The system shall allow administrators to configure the attendance compliance threshold percentage. Changes must take effect immediately and be logged with the administrator ID and timestamp. (BOR-1-WEB-01b; TR-1-BE-03, UC-06)
- **FR-OVS-03:** The system shall flag students whose attendance falls below the configured threshold. (BOR-1-WEB-01b, UC-04)
- **FR-OVS-04:** The system shall allow administrators to deliver a compliance warning letter to the registered parent email address of every student flagged below the attendance threshold. (BOR-1-WEB-01b, UC-17) [Delivery mechanism governed by [FR-SW-04.](#)]
- **FR-OVS-05:** The system must generate a WarningLetter record for each compliance warning letter sent, containing the student ID, threshold value breached, current attendance percentage, dispatch timestamp, and recipient email address. (BOR-1-WEB-01b, UC-17)
- **FR-OVS-06:** The system must save generated attendance reports for at least 12 months. Administrators cannot delete individual report records; the system archives reports on a regular basis. (BOR-0-02, BOR-0-03, UC-05, UC-08)

4.7 Leave and Medical Certificate Review

4.7.1 Description and Priority

- The system allows administrators to review leave submissions, validate medical certificates, and approve or reject leave requests. To meet Goal 4 and maintain attendance record integrity, administrators must review with MCS-backed validation
- Priority: High.
- Reference: BOR-1-WEB-02a and TR-1-BE-02, BOR-1-WEB-02b.

4.7.2 Stimulus/Response Sequences

- Authenticated administrators access the web portal's leave review screen.
- The system displays a list of pending leave requests, including student name, date(s), reason, and MCS validation status.



- The administrator selects a leave request and reviews the MCS validation result and associated documentation.
- The administrator approves or rejects the request, providing an optional comment.
- The system updates the attendance record and sends a push notification to the parent. If an administrator tries to approve a request with a failed MCS validation, the system will block the action and display an error.

4.7.3 Functional Requirements [UC-04, UC-07]

- **FR-LVR-01:** The system shall allow administrators to retrieve and view all pending leave requests, including their MCS validation outcome. (BOR-1-WEB-02a, UC-07)
- **FR-LVR-02:** The system shall prevent administrators from approving leave requests if the medical certificate has not been validated by the MCS. (BOR-1-WEB-02b; TR-1-BE-02, UC-07)
- **FR-LVR-03:** The system shall update the student's attendance record with the approved leave status upon administrator approval. (BOR-1-WEB-02a, UC-07)
- **FR-LVR-04:** The system shall update the status field of the LeaveRequest record to Approved or Rejected, while recording the administrator ID, decision timestamp, and any comments. (BOR-1-WEB-02a, BOR-1-WEB-02b, UC-07)
- **FR-LVR-05:** The system will not allow administrators to delete LeaveRequest records. All leave records must be kept in full for audit purposes. (BOR-0-03, BOR-1-WEB-02a)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The following performance requirements apply throughout the system. A rationale is provided for each to help guide design decisions. These requirements are based on TR-1-BE-04 (Revised), TR-1-BE-01b, and BioSec project Goals 1–3.

- **NFR-PERF-01:** The system shall load web portal pages within 3 seconds under normal operating conditions.
 - Rationale: Administrator productivity depends on responsive portal interactions for time-sensitive compliance and leave review actions. (TR-1-BE-01b)
- **NFR-PERF-02:** The system shall complete at least 95% of API responses within 500 milliseconds under normal load.
 - Rationale: API response latency affects the responsiveness of mobile applications for parents and teachers. (TR-1-BE-01b)
- **NFR-PERF-03:** The system shall generate attendance reports within 10 seconds for single class queries and within 60 seconds for institution-wide reports.
 - Rationale: Administrators need timely reporting to manage compliance actions on five campuses. (TR-1-BE-01b)
- **NFR-PERF-04:** The system shall deliver push notifications to a parent's device within 15 minutes of a missed session.
 - Rationale: Real-time parental notification within 15 minutes is a primary project goal (Goal 3) that parents must meet in order to take appropriate action. (BOR-1-MOB-04; TR-0-01)
- **NFR-PERF-05:** The system shall support at least 1,500 concurrent user sessions without compromising API response time or scan processing throughput.



- **Rationale:** The system must support at least 1,500 concurrent user sessions with graceful degradation to 2,000 users at a maximum 4-second response time. Auto-scaling shall trigger at 70% CPU utilisation. (TR-1-BE-01b)

5.2 Safety Requirements

Safety requirements address the potential harm caused by system failures, misuse, or data errors. BioSec operates across five campuses and processes biometric data from minors. Safety must be designed into the system architecture.

- **NFR-SAF-01:** The system shall prohibit overwriting or deleting attendance records, as uncontrolled data loss may result in incorrect attendance penalties for students. (TR-1-BE-03)
- **NFR-SAF-02:** Scanner failure shall not result in automatic absent marking for unscanned students, in order to prevent incorrect attendance penalties. (BOR-0-03; BOR-1-ATT-01) [Failure handling behaviour specified in [FR-HW-05](#).]
- **NFR-SAF-03:** The system shall encrypt all biometric data using AES-256 at rest and TLS 1.2 or higher in transit. Unencrypted exposure of biometric data constitutes a PDPA violation and may cause irreversible harm to individuals, as biometric data cannot be changed once compromised. (TR-1-BE-03; TR-1-BE-05b; TR-1-CPR-01)
- **NFR-SAF-04:** The system shall ensure audit logs are immutable once written. No user, including administrators, shall be permitted to modify or delete log entries, so as to prevent the concealment of incorrect or malicious attendance changes. (TR-1-BE-03)

5.3 Security Requirements

The security requirements protect biometric and personal data processed by the system. All requirements are consistent with Singapore's Personal Data Protection Act (PDPA) 2012, as amended. The system handles biometric data from minors, which necessitates the highest level of data security.

- **NFR-SEC-01:** The system shall encrypt biometric data, including fingerprint templates, at rest using AES-256 and in transit using TLS 1.2 or higher. No plaintext biometric data shall be stored or transmitted at any point. (TR-1-BE-03, TR-1-BE-05b)
- **NFR-SEC-02:** The system shall process and store all PII, including student records, parent contact information, and attendance data, in accordance with PDPA retention policy. (TR-1-BE-03)
- **NFR-SEC-03:** The system shall enforce two-factor authentication (2FA) for all administrator accounts accessing the web portal. (TR-1-BE-03; TR-1-CPR-01)
- **NFR-SEC-04:** The system shall lock all user accounts after a configurable number of failed login attempts, with a configurable lockout duration. All lockout events are logged with timestamp and account identifier. (TR-1-CPR-01)
- **NFR-SEC-05:** The system shall conduct all MCS API communications using HTTPS and OAuth 2.0 authentication. No other authentication mechanism shall be permitted. (TR-1-BE-02)
- **NFR-SEC-06:** The system shall retain audit logs, including login events, attendance changes, leave approvals, and access denials, for at least 12 months and shall protect them from modification or deletion. (TR-1-CPR-01)
- **NFR-SEC-07:** The system shall restrict web portal access to campus networks or secure VPN connections. Direct internet access to the administrator portal shall not be permitted. (TR-1-CL-02)
- **NFR-SEC-08:** The system shall require mobile applications to be code-signed with BioSec institutional certificates before distribution on Google Play and Apple App Store. (TR-1-CL-01d)
- **NFR-SEC-09:** The system shall encrypt all locally cached data on mobile devices using AES-256. Data cached for offline use shall be automatically purged after 48 hours. (TR-1-CL-01f)



5.4 Software Quality Attributes

The following quality attributes are defined using McCall's Quality Model (1977), as applied to Nevon Solutions' software development quality standard. McCall's model divides quality into three dimensions: operation (how well the software runs), revision (how well it tolerates internal change), and transition (how well it adapts to external change). Eleven quality features are chosen and applied to the BioSec system shown below.

5.4.1 Operation Perspective

- **QA-CORRECT-01:** The system shall ensure that attendance records accurately reflect biometric scan results without discrepancies. When the same student record is referenced concurrently in attendance, leave, and notification tables, the values must be consistent across all data contexts.
 - Verification: Automated reconciliation tests for the AttendanceRecord, LeaveRequest, and NotificationLog tables.
- **QA-EFFIC-01:** The system shall process biometric scans in 2 seconds and respond to API requests in 500ms for 95% of calls, using the minimum Azure configuration specified in TR-1-BE-01a and TR-1-BE-01b. No redundant database queries will be executed per user transaction.
 - Verification: Load testing with 1,500 concurrent users and 200 concurrent scans.
- **QA-INTEG-01:** The system shall prevent unauthorized access to biometric data, attendance records, and system configuration using RBAC, AES-256 encryption, TLS 1.2+, 2FA for administrators, and account lockout. Any unauthorized access attempt will be recorded.
 - Verification includes penetration testing and a PDPA compliance audit prior to go-live.
- **QA-RELIA-01:** The system shall maintain 99.9% availability during BioSec's operational hours for 30 days. Failures in scanner-to-backend data transmission do not result in silent data loss; instead, failed transmissions are queued for retry.
 - Verification: 14-day uptime verification during soft launch based on delivery acceptance criteria.
- **QA-USAB-01:** The system shall allow parent and teacher users must be able to complete primary tasks (view attendance, submit leave requests) within three interactions from the home screen, without referring to the user manual. The interface must accommodate users with varying technical literacy.
 - Verification: Entails conducting usability tests with representative parent and teacher participants before deployment.

5.4.2 Revision Perspective

- **QA-FLEX-01:** The system shall allow administrators to configure the attendance compliance threshold percentage without requiring software changes or redeployment. Push notification delivery channels (Firebase, APNS, and SMS fallback) must be configurable without requiring code changes.
 - Verification: Configuration change tests in the staging environment.
- **QA-MAINT-01:** The system operational issues shall be classified using a three-tier severity model: Critical (4-hour response, 24-hour resolution), High (8-hour response, 48-hour resolution), and Medium (24-hour response, 5-day resolution) during the 12-month warranty period. (TR-1-CPR-04)
- **QA-TEST-01:** The system shall generate structured audit logs sufficient to reproduce and diagnose any reported incident. API endpoints must comply with the OpenAPI specification to allow for automated integration testing. (TR-1-DOC-02)



5.4.3 Transition Perspective

- **QA-INTEROP-01:** The system shall exchange data with the MCS using HTTPS REST and OAuth 2.0, and with Firebase and APNS using their standard APIs. All external integrations must use documented and versioned APIs. Integration failures must not affect existing attendance or leave records. (TR-1-BE-02, Section 3.3)
- **QA-PORT-01:** Mobile applications must be distributed as native Android (minimum Android 10) and iOS (minimum iOS 14) apps. The backend must be deployable on Microsoft Azure without platform-specific code changes. (TR-0-01, TR-0-02, TR-1-CL-01a, TR-1-CL-01b)
- **QA-REUSE-01:** The RESTful API layer must be designed so that future BioSec applications can consume attendance and student data endpoints without requiring changes to the backend. API documentation should be delivered in Swagger/OpenAPI format. (TR-1-DOC-03; TR-1-DOC-04)

5.5 Business Rules

The following operating principles determine which roles can perform which functions and under what conditions. These are not functional requirements, but rather functional constraints imposed by the system.

- **BR-01:** Only biometric scan results are considered valid attendance records. Manual attendance recording is not permitted except through the supervised manual override flow as specified in [FR-HW-04](#). (BOR-0-03; BOR-1-ATT-01)
- **BR-02:** Attendance capture only permitted during officially scheduled academic sessions. Scans outside session windows rejected. (BOR-1-ATT-01)
- **BR-03:** Administrators will not approve leave requests with a medical certificate unless MCS validates successfully. (BOR-1-WEB-02b; BOR-1-MOB-03b)
- **BR-04:** Parents can only view their registered child's attendance. (BOR-1-MOB-02a; TR-1-CPR-01)
- **BR-05:** Teachers can only view assigned classes. (BOR-1-MOB-02b; TR-1-CPR-01)
- **BR-06:** Leave submissions limited to attendance-related absences. The system excludes leave for academic grading, appointments, and complaints. (BOR-1-MOB-03a)
- **BR-07:** The system will automatically send compliance warning letters when attendance falls below threshold. Administrators cannot prevent a letter from being sent. (BOR-1-WEB-01b)
- **BR-08:** Biometric data shall only be used for attendance purposes. Not shared with third parties without explicit legal permission. (PDPA; TR-1-BE-03)
- **BR-09:** Audit logs are the official record of all system events. No user role may modify or delete entries. (TR-1-BE-03)

6. Other Requirements

This section covers requirements not covered in Sections 4 and 5, such as database, legal, data migration, documentation, and maintenance needs.

6.1 Database Requirements

- **NFR-DB-01:** The system shall use Microsoft SQL Server hosted on Azure SQL Database. TR-1-DOC-02 requires the delivery of a complete database schema and stored procedure documentation. Daily automated backups will be performed, with monthly restoration verification. (TR-1-BE-05a, TR-1-DOC-02, TR-1-BE-05c)
- **NFR-DB-02:** The system's database schema shall ensure referential integrity, preventing attendance records, leave requests, and notifications from referencing non-existent student or user



accounts. Duplicate student IDs will be detected and rejected at both the application and database layers. (TR-1-BE-05a, TR-1-CPR-03)

- **NFR-DB-03:** All biometric template data in the database to be encrypted with AES-256. No plaintext biometric data shall be written to the database at any time. The format for storing biometric templates is an open issue that has yet to be resolved. (TR-1-BE-05b)

6.2 Data Migration Requirements

- **NFR-MIG-01:** The system shall support migration of legacy attendance data from Excel and CSV formats before going live. All current academic year attendance data must be migrated and validated before the system goes into production. (TR-1-CPR-03)
- **NFR-MIG-02:** The system shall detect and flag duplicate student IDs during data migration. Duplicate records will not be imported and must be reported to BioSec for resolution before the migration is marked as complete. (TR-1-CPR-03)

6.3 Legal and Regulatory Requirements

- **NFR-LEGAL-01:** The system shall adhere to Singapore's Personal Data Protection Act (PDPA) 2012 as amended when collecting, using, disclosing, storing, and disposing of personal and biometric data. BioSec retains institutional data ownership for all attendance and biometric records (TR-1-BE-03, PDPA).
- **NFR-LEGAL-02:** BioSec will own all source code, database schemas, API documentation, user manuals, and system architecture artifacts delivered during the project. TR-1-DOC-01, TR-1-DOC-02, TR-1-DOC-03, and TR-1-DOC-04.

6.4 Maintenance and Support Requirements

- **NFR-MAINT-01:** Support for new Android and iOS operating system versions to be delivered within 30 days of public release during warranty period. During the warranty period, all application version updates must be backward compatible. (TR-1-CPR-04; TR-1-CL-01e)

6.5 Document Requirements

- **NFR-DOC-01:** A PDF version of the Administrator, Teacher, and Parent User Manuals delivered with the system. Each manual must include screenshots and campus-specific procedures. (TR-1-DOC-04)
- **NFR-DOC-02:** API documentation in Swagger/OpenAPI format, including endpoints, request and response formats, authentication, and error codes. Database schema documentation, including ERD and stored procedures, must also be delivered. (TR-1-DOC-02, TR-1-DOC-03)
- **NFR-DOC-03:** Comprehensive system architecture diagrams and deployment documentation to support future campus expansion and disaster recovery planning. Disaster recovery RPO/RTO specifications are still pending BioSec approval. (TR-1-DOC-04, TR-1-DOC-01)

7. Inceptive Requirements Traceability Matrix (iRTM)

The Inceptive Requirements Traceability Matrix (iRTM) is the first segment of the Requirements Traceability Matrix (RTM) for this project. It systematically establishes the chain of custody for all requirements from BioSec Project Specification (BS-PS-2026-001) to Nevon Solutions Software Requirements Specification (NS-SRS-2026-001), ensuring that every requirement elicited is accounted for in the specification.



The iRTM is configured as a table using the standard categories "Biz Ops" and "IT Tech" for the PS requirements. Each row traces a PS requirement to its corresponding SRS requirement(s), with the cardinality of PS to SRS clauses being any combination of 1..1, 1..many, many..1, and many..many. The matrix includes the following fields for each side: requirement category, unique codified identifier, description (as paraphrases or minimizations of the original clauses), stakeholder or owner role, and precedence or ranking.

As a living document, the iRTM will be horizontally right-appended in subsequent SDLC stages to track the fulfilment of requirements through to deployment. Any changes to requirements governed by the Change Mitigation strategies defined in Section 8 shall be reflected in the iRTM to maintain traceability integrity.

The complete iRTM is provided in [Appendix D](#).

8. Change Mitigation Strategies

Change to SRS content is inevitable due to stakeholder changes of mind, industry shifts, and government regulation updates. The following three strategies are devised to protect Nevon Solutions' project profitability and schedule integrity while maintaining goodwill with BioSec. Each strategy is codified with the CM- prefix.

8.1 CM#1: Anticipatory Contingency Requirements

These are requirements pre-approved by stakeholders and placed in the SRS, to be activated only when specific triggering conditions or events occur.

CM-CON-01: If Singapore PDPA amendments affecting biometric data handling are enacted after SRS release, the system shall defer implementation of affected data retention and consent workflows until updated regulatory guidance is published. During deferral, the system shall enforce the most restrictive interpretation of existing PDPA provisions as a temporary safeguard. (Rationale: PDPA amendments are under active review by PDPC.)

CM-CON-02: If campus access policies are modified due to organizational restructuring at BioSec, a temporary policy override module shall be activated allowing administrators to configure campus-specific attendance rules for a period not exceeding 60 days, after which the permanent policy framework must be updated. (Rationale: 5 campuses with potential for policy divergence.)

CM-CON-03: If the MCS vendor deprecates the current REST API version (v2.x), the system shall activate an adapter layer supporting both legacy and new API endpoints concurrently for a transition period of 90 days. MC verifications shall continue via the legacy endpoint with degraded response time tolerance increased from 5s to 10s. (Rationale: MCS API stability depends on third-party vendor roadmap.)

8.2 CM#2: Open Issues

These are KIV requirements that could not be agreed upon during the Analysing phase. They are placed in the SRS with the purpose of continuing negotiations after development has commenced. Each has a kill date; if not resolved by that date, the fallback action takes effect.

ID	Issue	Description & Fallback	Kill Date	Origin
CM-OI-01	MCS Integration Security	Encryption handshake protocol for mutual TLS with MCS vendor pending. Fallback: Standard TLS 1.2 with OAuth 2.0 bearer tokens.	20 Mar 2026	AR Section 3.5, Issue #1



CM-OI-02	Disaster Recovery RPO/RTO	BioSec has not confirmed RPO/RTO values. Fallback: Default RPO 1 hour, RTO 4 hours.	20 Mar 2026	<i>AR Section 3.5, Issue #2</i>
CM-OI-03	Biometric Template Format	ISO/IEC 19794-2 vs proprietary format decision pending. Fallback: ISO/IEC 19794-2 as default for interoperability.	15 Mar 2026	<i>AR Section 3.5, Issue #3</i>

Table 2: CM#2 Open Issues with Kill Dates

8.3 CM#3: High-Risk Change Control Process

This strategy establishes a pre-approved process for changes to requirements designated as high-risk (specifically impactful to large parts of the system).

Designated High-Risk Requirements: NFR-SEC-01 (biometric encryption), FR-HW-02 (scan processing), NFR-PERF-01 (response time), FR-NOTIF-01 (notification delivery), NFR-LEGAL-01 (PDPA compliance).

Change Control Process:

- **Step 1 - Submission:** BioSec submits a formal Change Request using the template provided in [Appendix E](#), specifying the requirement ID, reason, proposed modification, and expected business value.
- **Step 2 - Impact Assessment:** Nevon Solutions conducts an impact assessment within 5 working days covering technical feasibility, schedule impact, budget impact, and dependent requirements.
- **Step 3 - Review:** A joint Change Control Board (CCB), comprising at least one designated representative from BioSec and one from Nevon Solutions, reviews the CR and impact assessment, prioritizing by criticality and substantiated business value. Quorum requires both parties to be present.
- **Step 4 - Approval:** If approved, the CCB negotiates schedule extensions and, for very high-risk requirements (NFR-SEC-03, NFR-LEGAL-01), contract amendments to adjust budget.
- **Step 5 - Implementation:** The change is incorporated into the SRS and the iRTM is updated to reflect the modification chain.

Constraint: Change requests are valid only for requirements that already exist in this SRS. Requests for new requirements not present in the SRS are deemed invalid and dropped forthwith.



9. SRS Sign off

This Software Requirements Specification (NS-SRS-2026-001) has been reviewed and is submitted for approval. The document defines the agreed baseline of functional and non-functional requirements for the Biometric Student Attendance System, and shall serve as the authoritative reference for all subsequent design, development, testing, and deployment activities.

By signing below, the Project Lead certifies the following:

- The Software Requirements Specification (Version 0.8) has been prepared in accordance with the project's requirements engineering methodology and is internally consistent.
- All requirements derived from the Project Specification (BS-PS-2026-001) have been accounted for through the Analysing process and are traceable via the Inceptive Requirements Traceability Matrix (iRTM).
- The documented requirements constitute the approved development baseline, inclusive of open issues and change mitigation strategies defined herein.
- All modifications to this baseline after sign-off shall be governed exclusively by the Change Mitigation Strategies specified in Section 8 of this document.

Vendor Authenticator

 <u>Loh Wen Xuan</u> Name (Project Lead)	 Signature	<u>06/03/2026</u> Date
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Designated Approvers – Vendor (Nevon Solutions)

Role	Name	Signature	Date
Technical Lead	James Koh		06/03/2026
System Architect Lead	Rachel Lee		06/03/2026

Designated Approvers – Customer (BioSec)

Role	Name	Signature	Date
Project Manager	Marcus Tan Wei Liang		
Head of Operations	Daniel Rodriguez		





Figures

The following models were developed during the Eliciting (PS) and Analysing phases and are brought into the SRS as required.

Figure 1: Context Data Flow Diagram

The Context DFD (revised during TC-02 clarification session, 9 February 2026) shows the Biometric Student Attendance System and its interactions with all external entities.

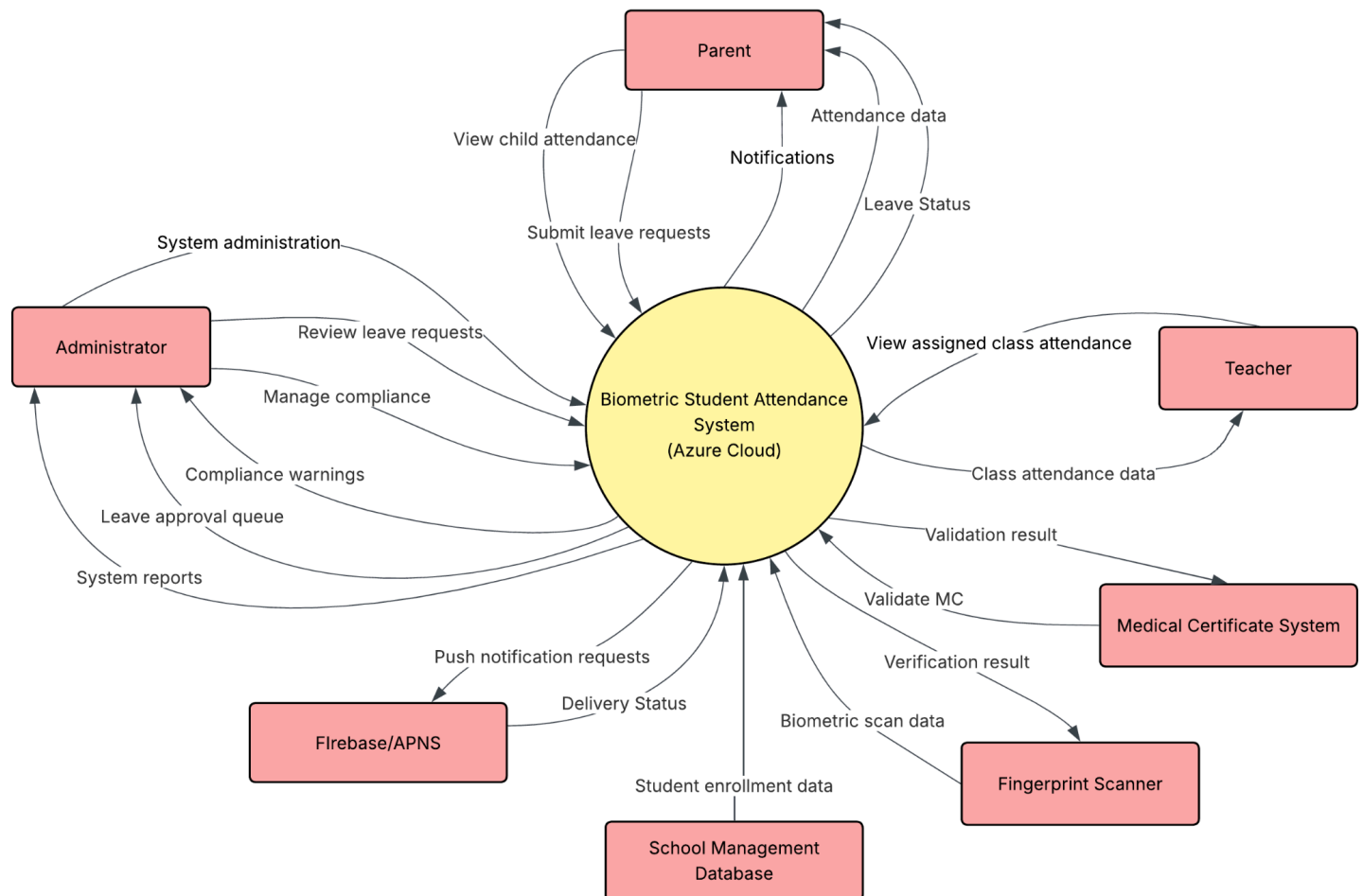


Figure 1: Context Data Flow Diagram (revised TC-02, 09/02/2026)



Figure 2 Wireframes

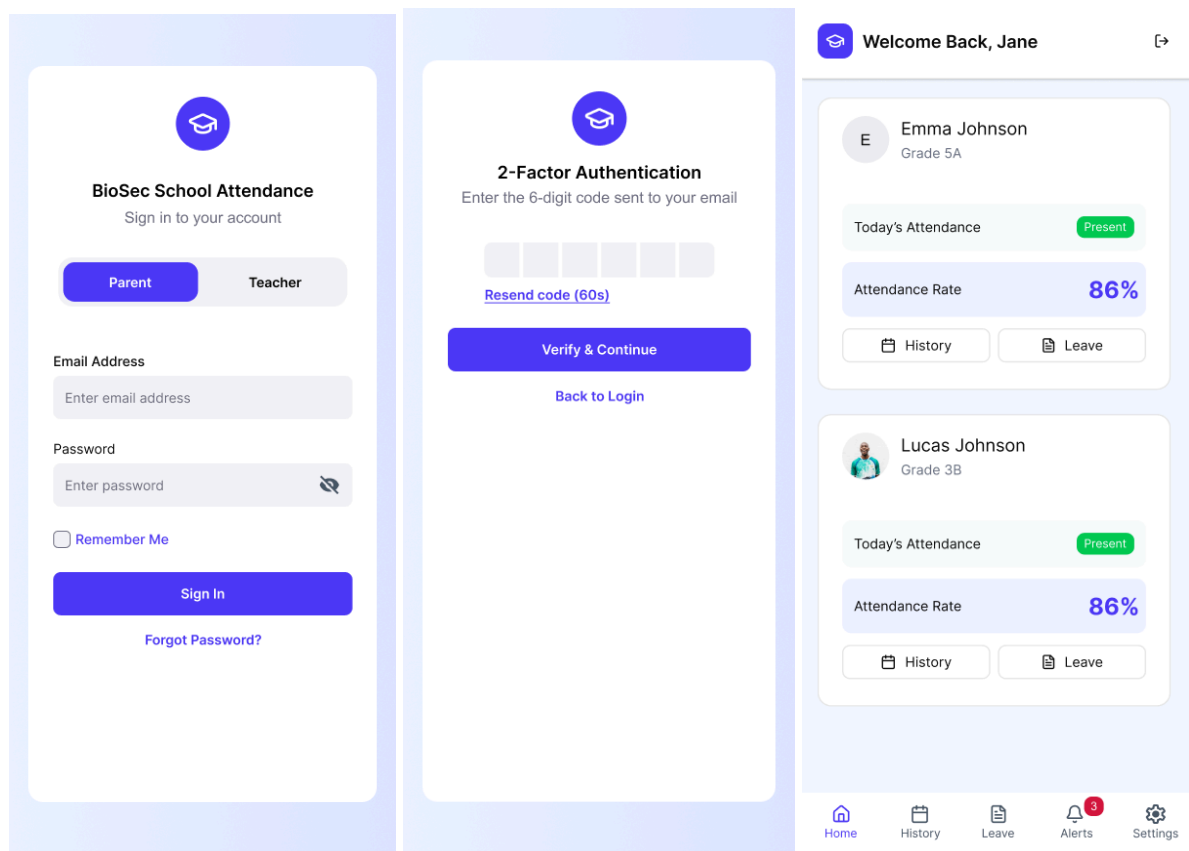


Figure 2.1: Parent/Teacher Login Screen — BioSec School Attendance (UR-GUI-02, 07/02/2026)

Figure 2.2: Parent/Teacher 2-Factor Authentication Screen — BioSec School Attendance (UR-MOB-06, 07/02/2026)

Figure 2.3: Parent Home Screen — BioSec School Attendance (UR-MOB-01, 07/02/2026)

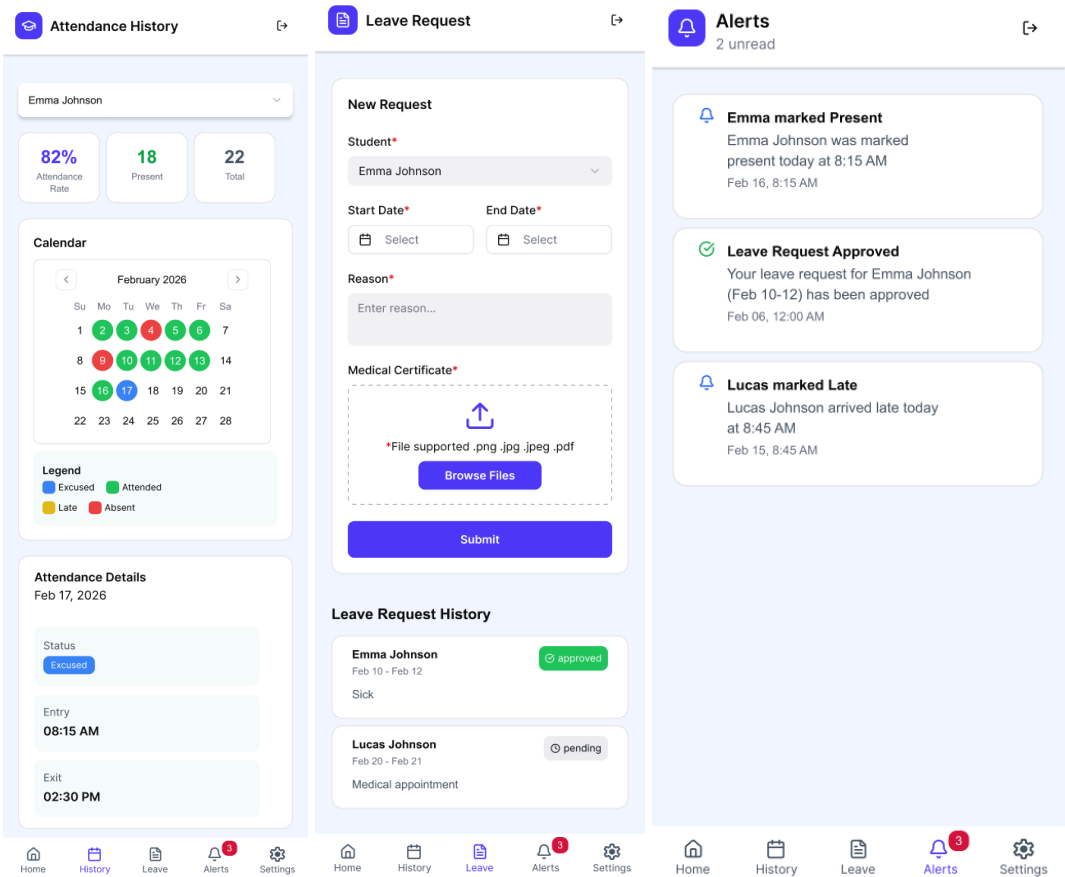


Figure 2.4: Parent Attendance History Screen — BioSec School Attendance (UR-MOB-01, 07/02/2026)

Figure 2.5: Parent Leave Request and Leave History Screen — BioSec School Attendance (UR-MOB-02, 07/02/2026)

Figure 2.6: Parent Alert Screen — BioSec School Attendance (UR-MOB-04, 07/02/2026)

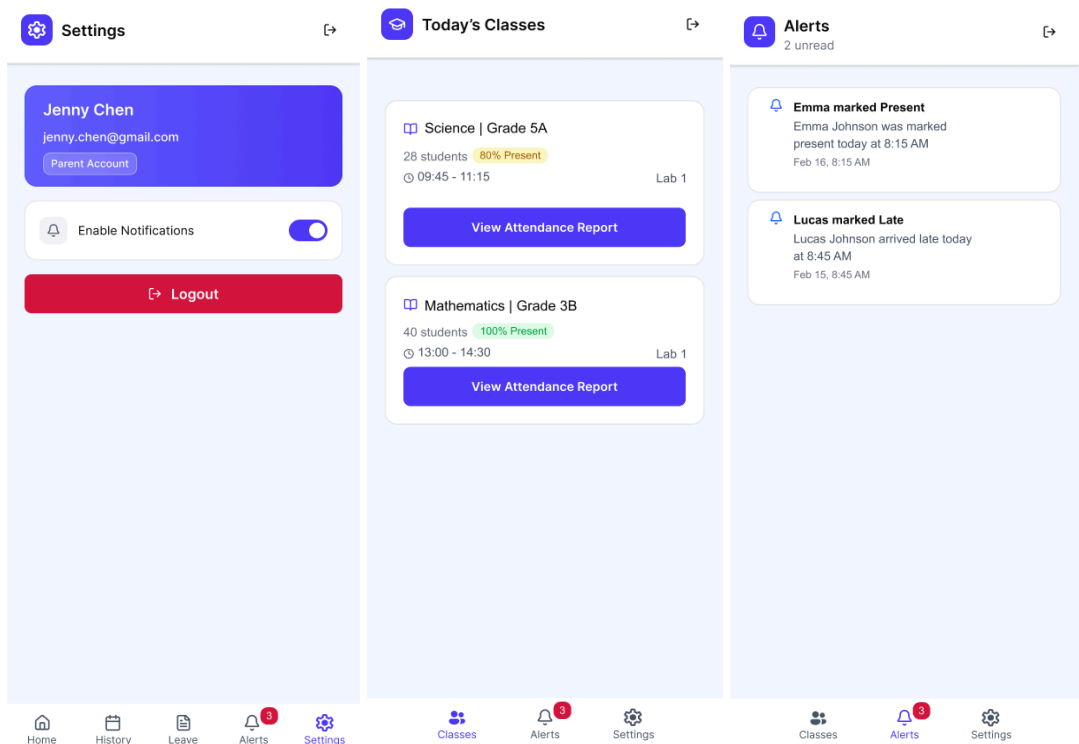


Figure 2.7: Setting Screen — BioSec School Attendance (BC-01, 07/02/2026)
Figure 2.8: Teacher Home Screen — BioSec School Attendance (UR-MOB-03, 07/02/2026)
Figure 2.9: Teacher Alert Screen — BioSec School Attendance (UR-MOB-04, 07/02/2026)

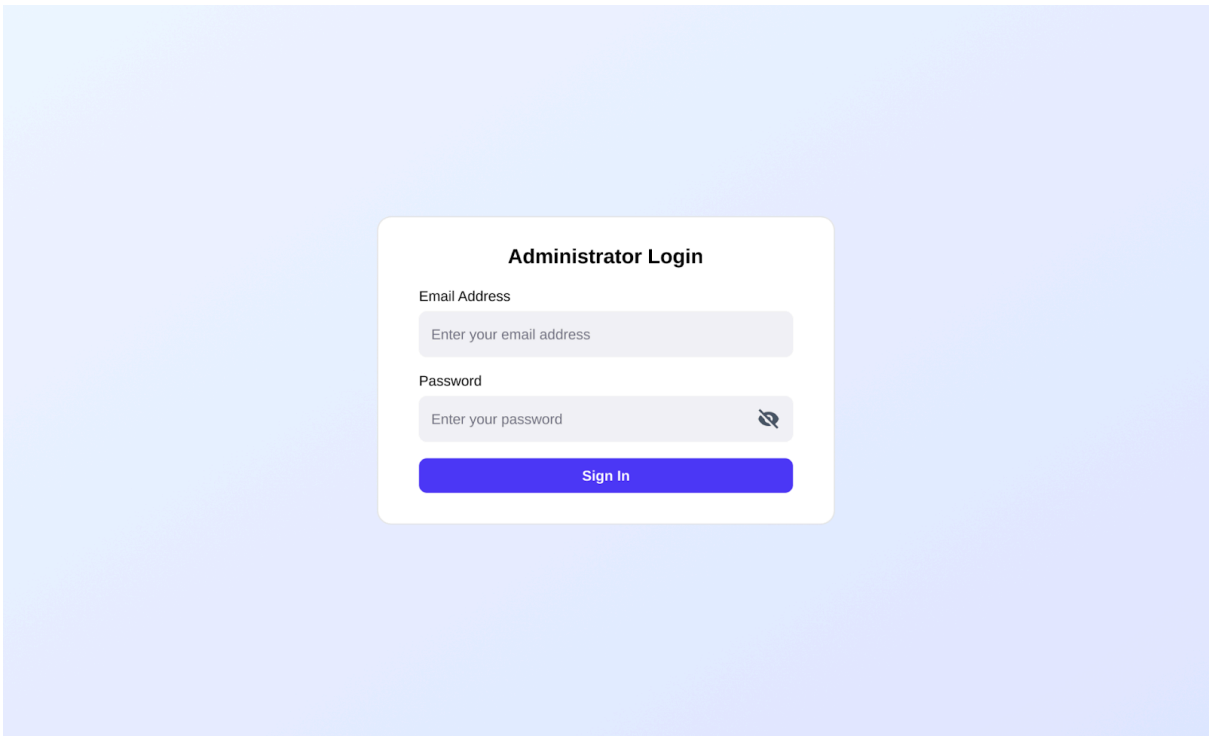


Figure 2.10: Administrator Login Screen — BioSec School Attendance (UR-GUI-02, 07/02/2026)

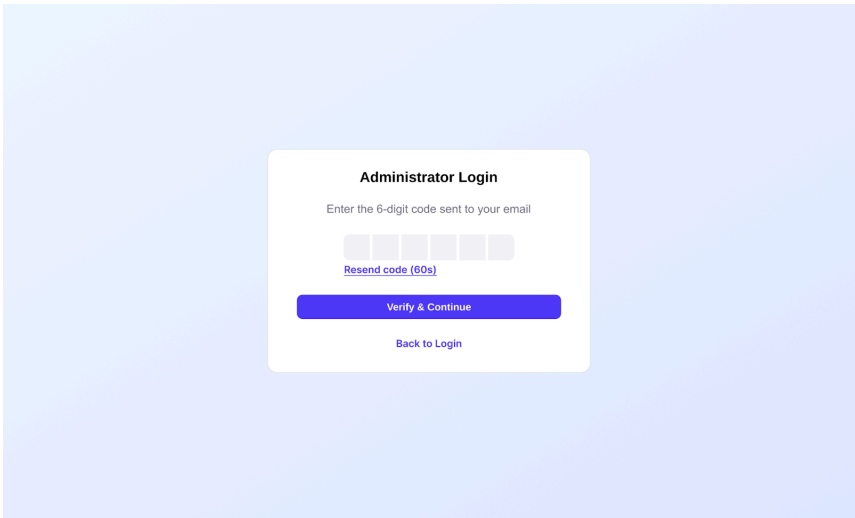


Figure 2.11: Administrator 2-Factor Authentication Screen — BioSec School Attendance (UR-WEB-09, 07/02/2026)

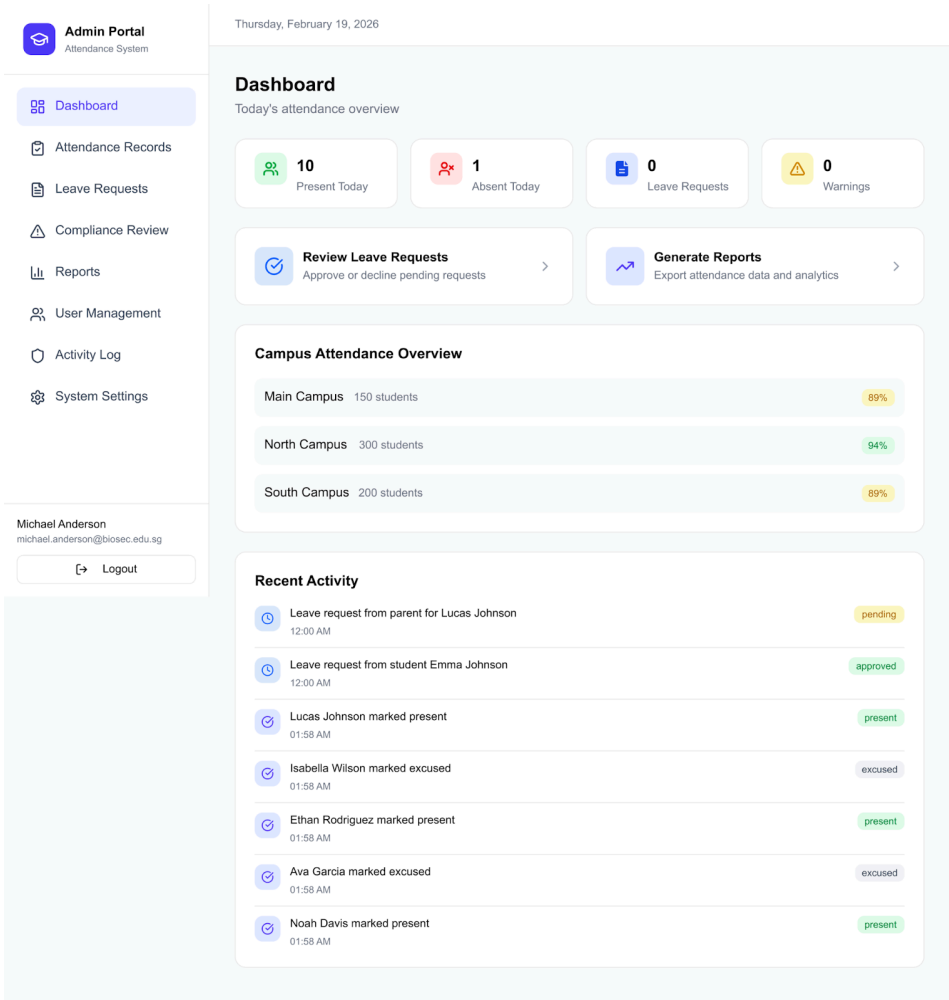


Figure 2.12: Administrator Dashboard Screen — BioSec School Attendance (UR-WEB-01, 07/02/2026)



**Admin Portal**
Attendance System

Dashboard

Attendance Records

Leave Requests

Compliance Review

Reports

User Management

Activity Log

System Settings

Michael Anderson
michael.anderson@biosec.edu.sg

Logout

Thursday, February 19, 2026

Attendance Records

156 records found

Export

Start Date

End Date

Campus

Class

Status

Jan 21, 2026

Feb 21, 2026

All Campuses

All Classes

All Statuses

Student Name

Search student...

1 student selected

Mark Manually

☐	Student Name	Class	Status	Scan Time	Campus
☒	Emma Johnson	Grade 5A	Present	Feb 18, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Present	Feb 18, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 17, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Excused	Feb 17, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 16, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Present	Feb 16, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 13, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Present	Feb 13, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 12, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Present	Feb 12, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Absent	Feb 11, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Present	Feb 11, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 10, 02:06 AM	Main Campus
☐	Lucas Johnson	Grade 3B	Absent	Feb 10, 02:06 AM	Main Campus
☐	Emma Johnson	Grade 5A	Present	Feb 9, 02:06 AM	Main Campus

Figure 2.13: Administrator Attendance Record Screen — BioSec School Attendance (UR-WEB-01, 07/02/2026)

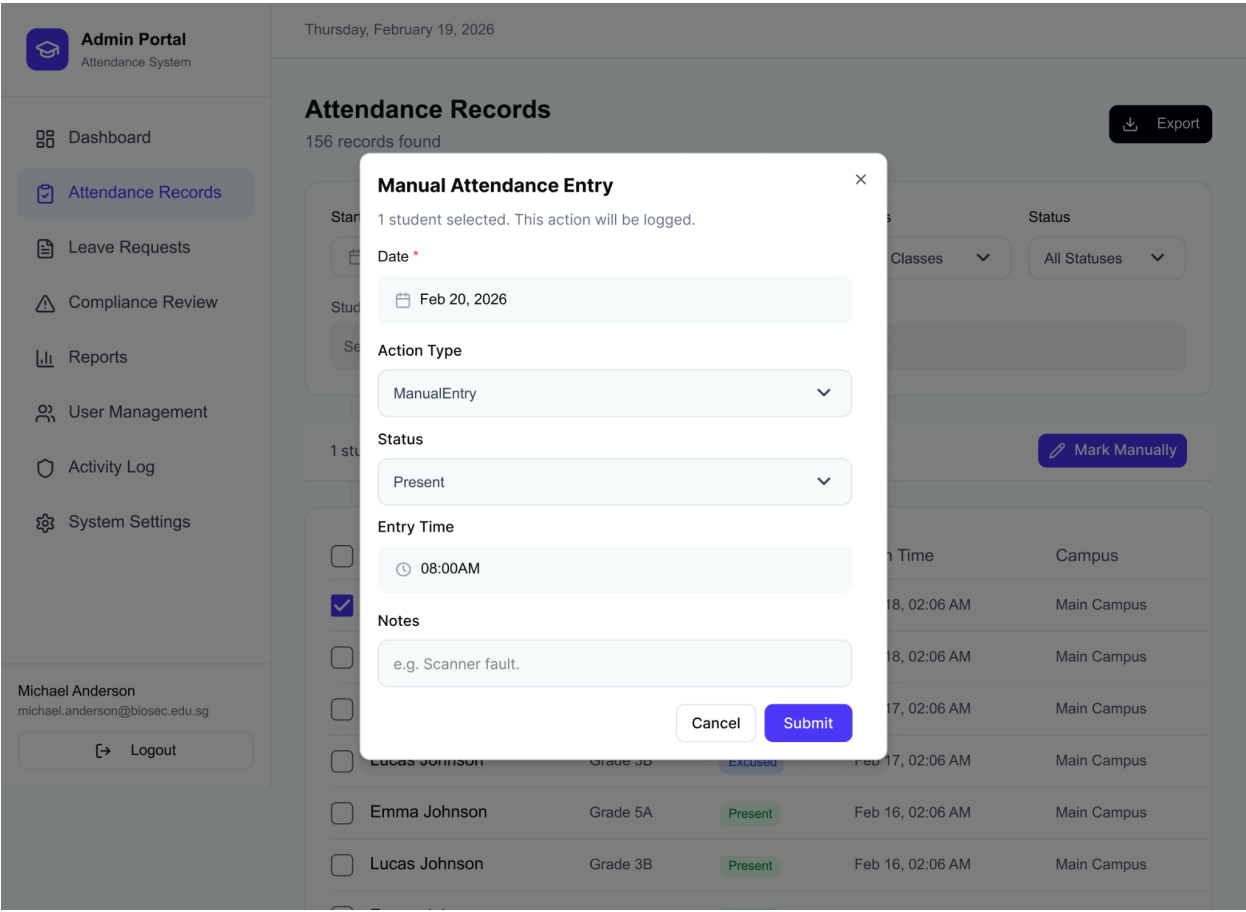


Figure 2.14: Administrator Manual Attendance Entry Screen — BioSec School Attendance (UR-WEB-10, 07/02/2026)

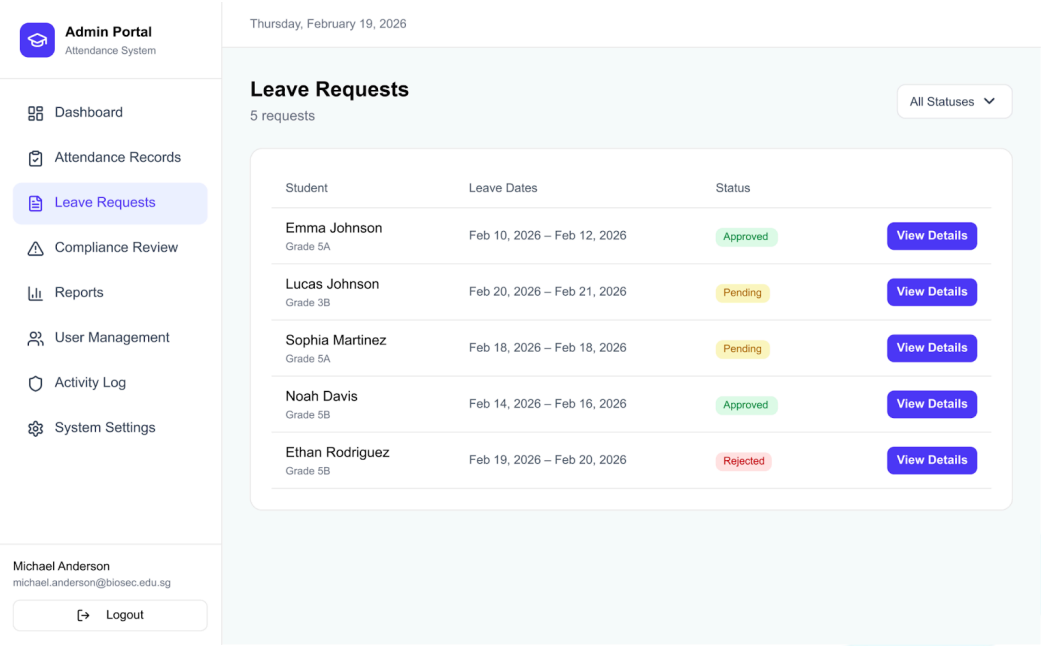


Figure 2.15: Administrator Leave Request Screen — BioSec School Attendance (UR-WEB-02, 07/02/2026)

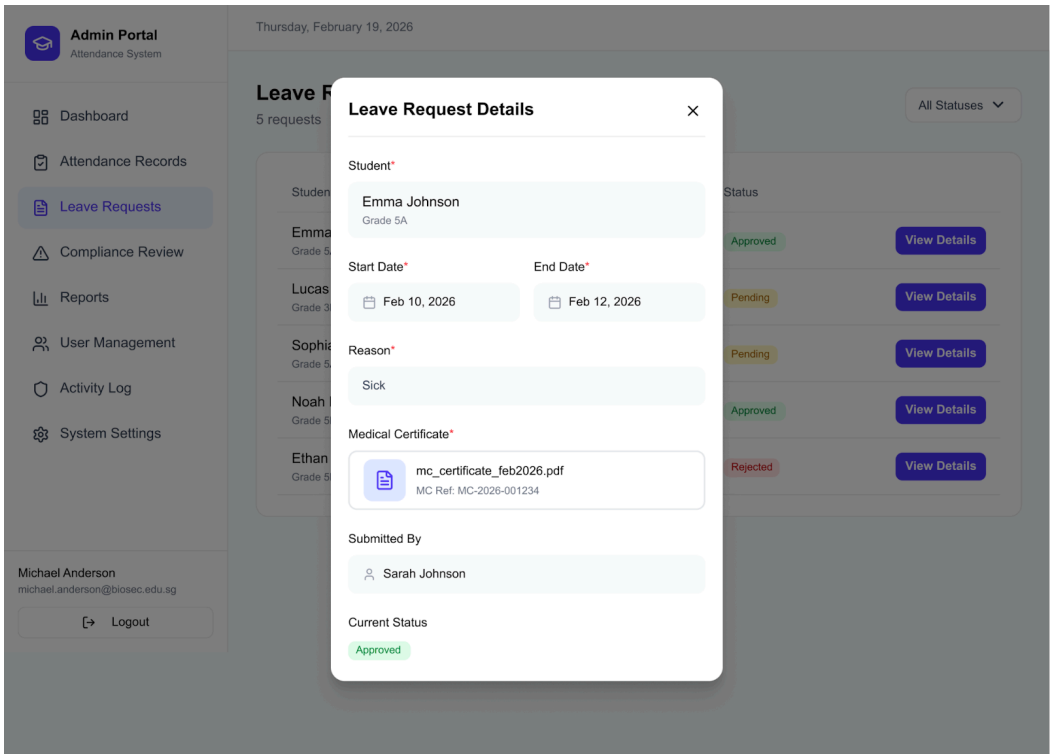


Figure 2.16: Administrator View Leave Request Screen — BioSec School Attendance (UR-WEB-02, 07/02/2026)

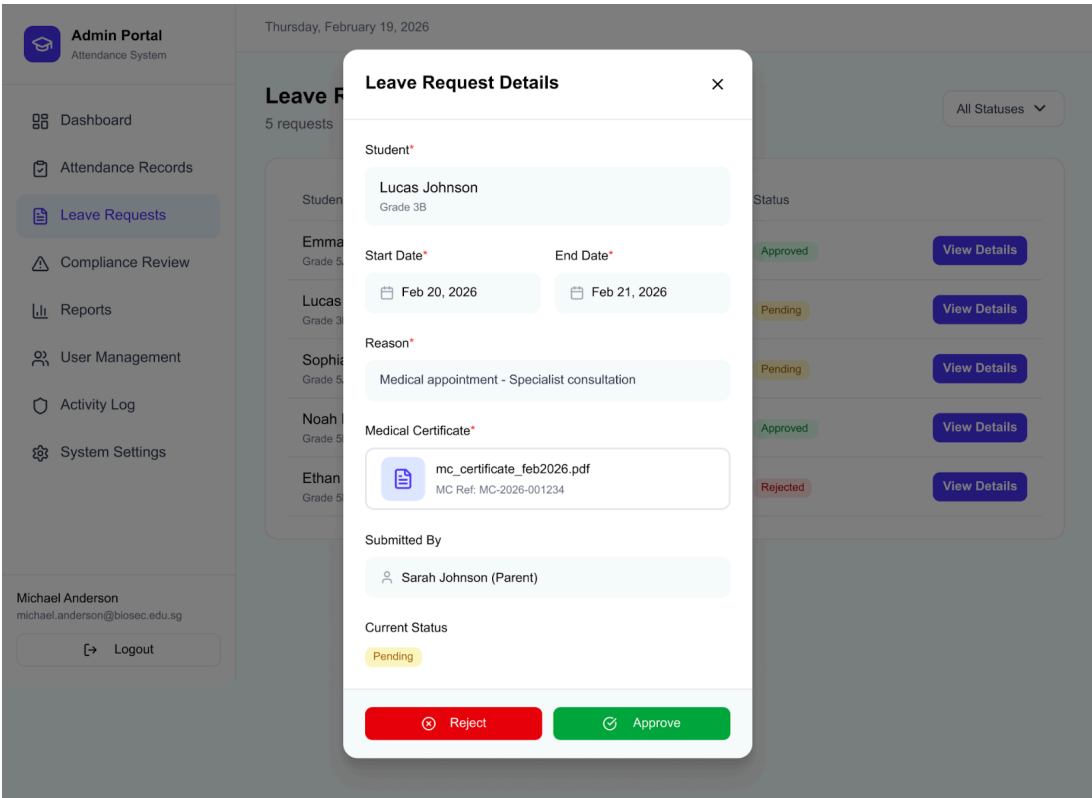


Figure 2.17: Administrator Approve Leave Request Screen — BioSec School Attendance (UR-WEB-02, 07/02/2026)

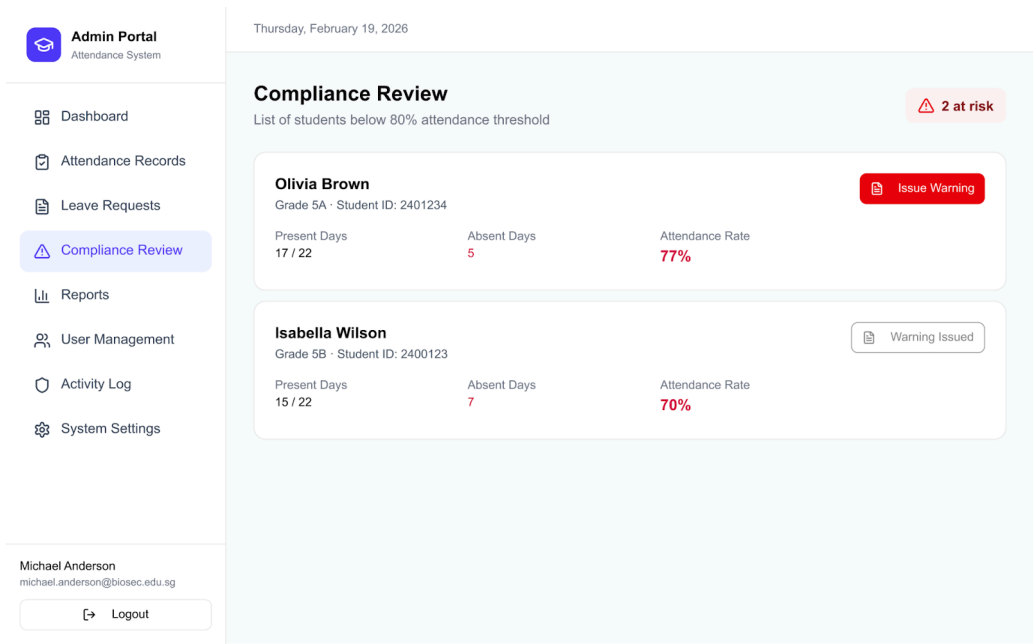


Figure 2.18: Administrator Compliance Review Screen — BioSec School Attendance (UR-WEB-03 , 07/02/2026)

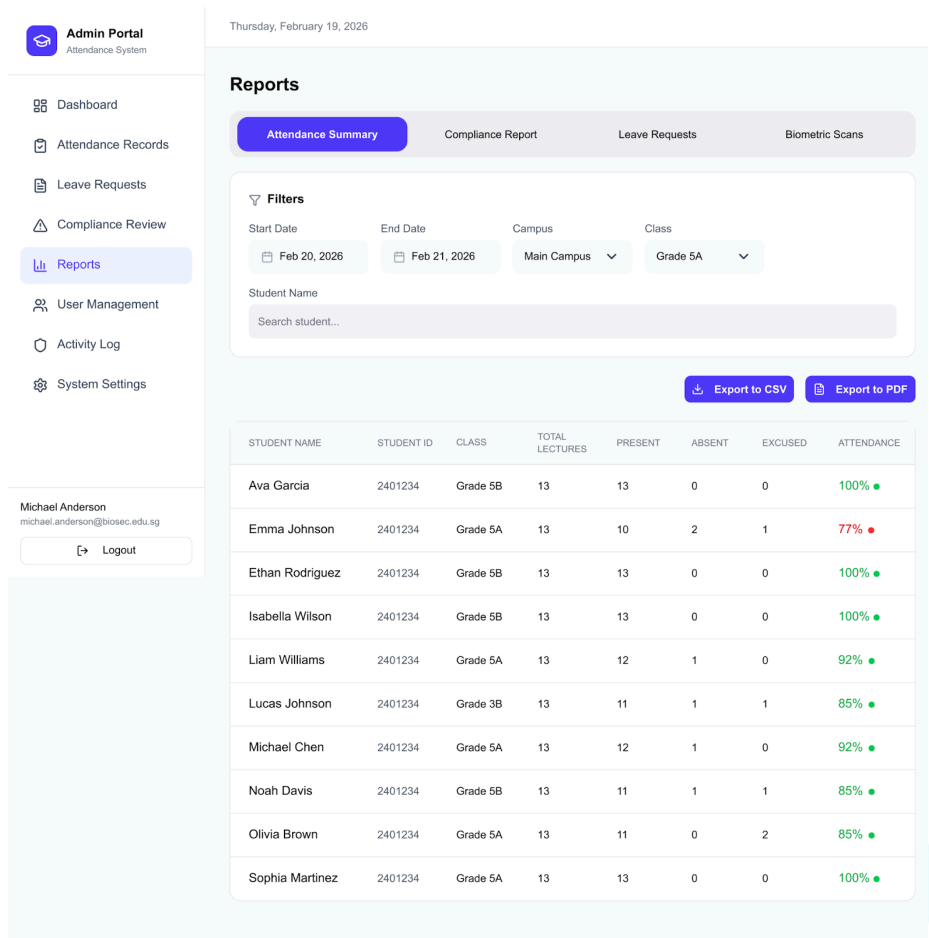


Figure 2.19: Administrator Attendance Summary Screen — BioSec School Attendance (UR-WEB-05, 07/02/2026)

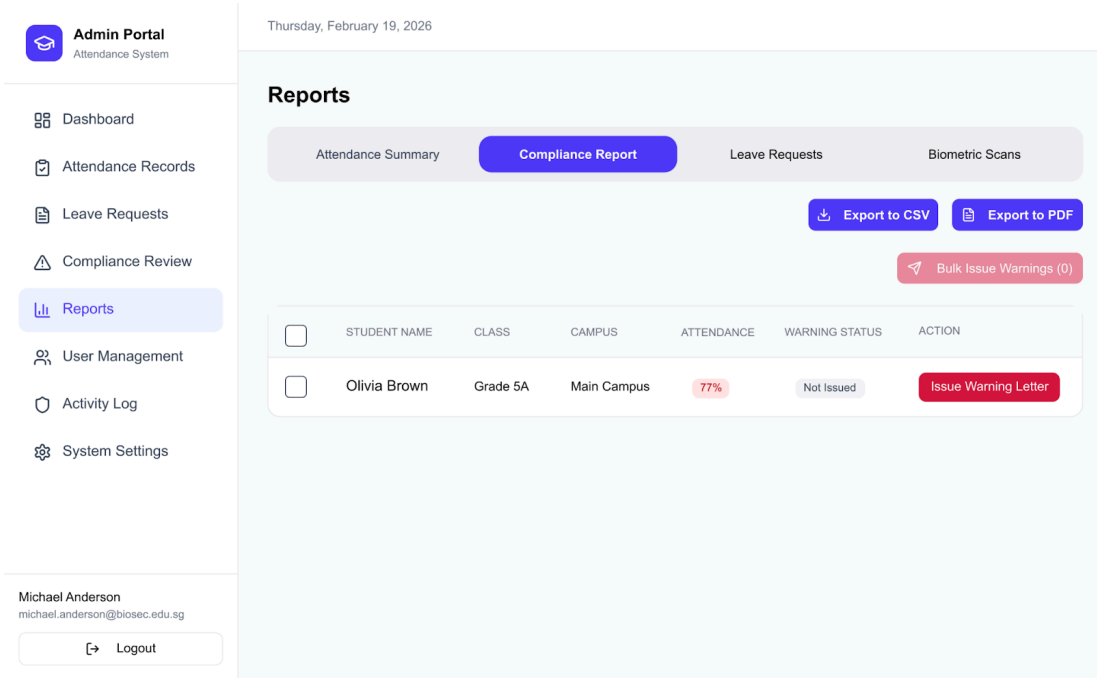


Figure 2.20: Administrator Compliance Report Screen — BioSec School Attendance (UR-WEB-05, 07/02/2026)

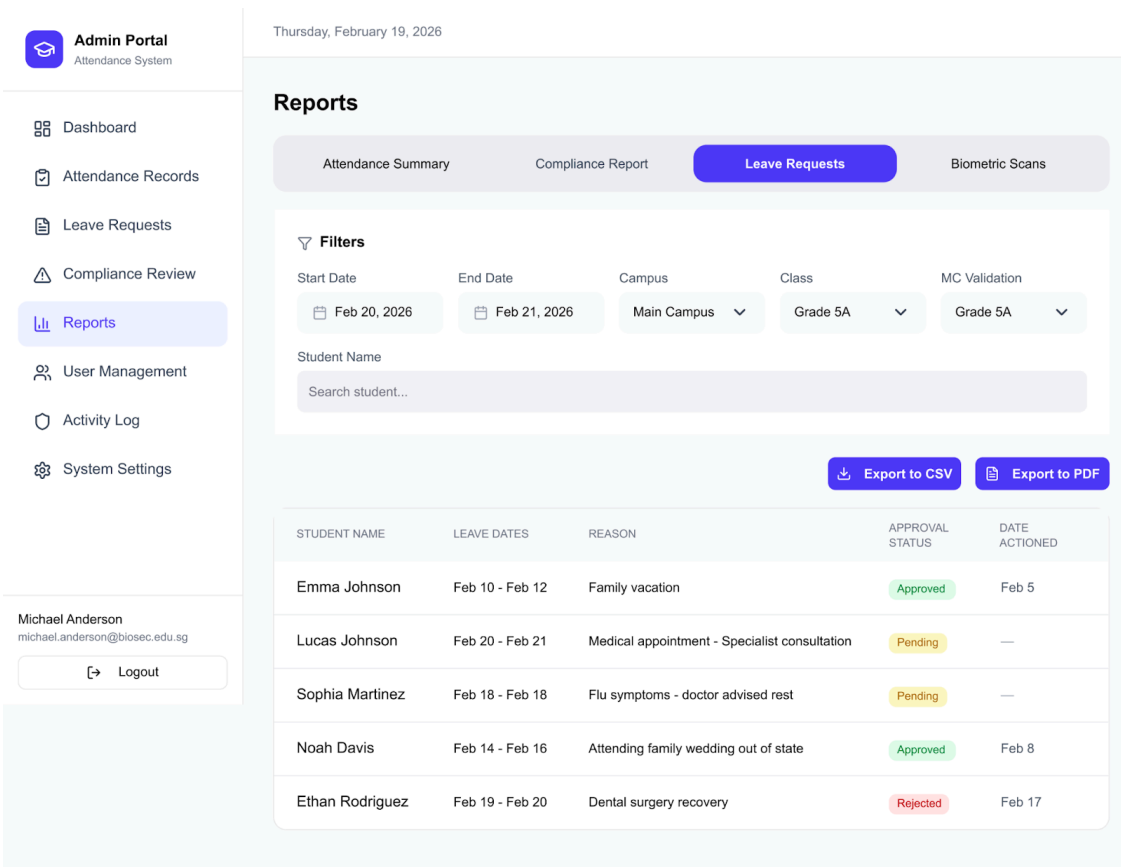


Figure 2.21: Administrator Leave Request Report Screen — BioSec School Attendance (UR-WEB-05, 07/02/2026)

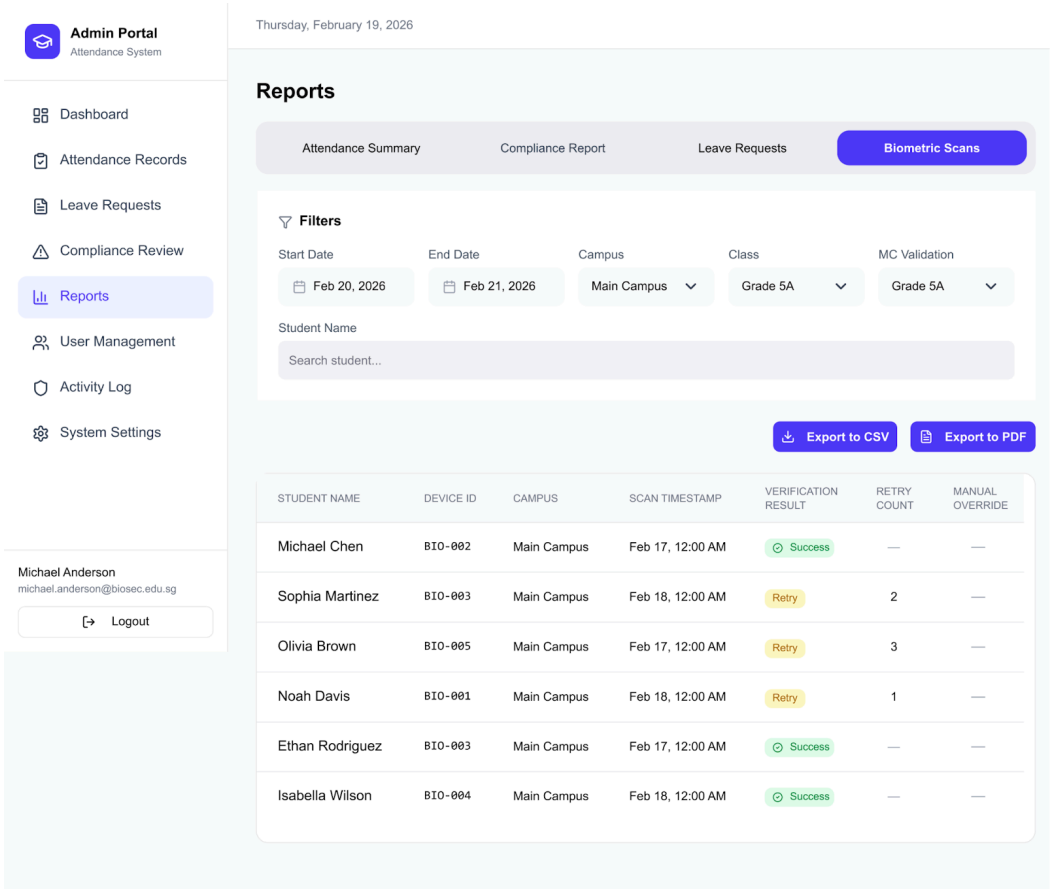


Figure 2.22: Administrator Biometric Scans Report Screen — BioSec School Attendance (UR-WEB-05, 07/02/2026)

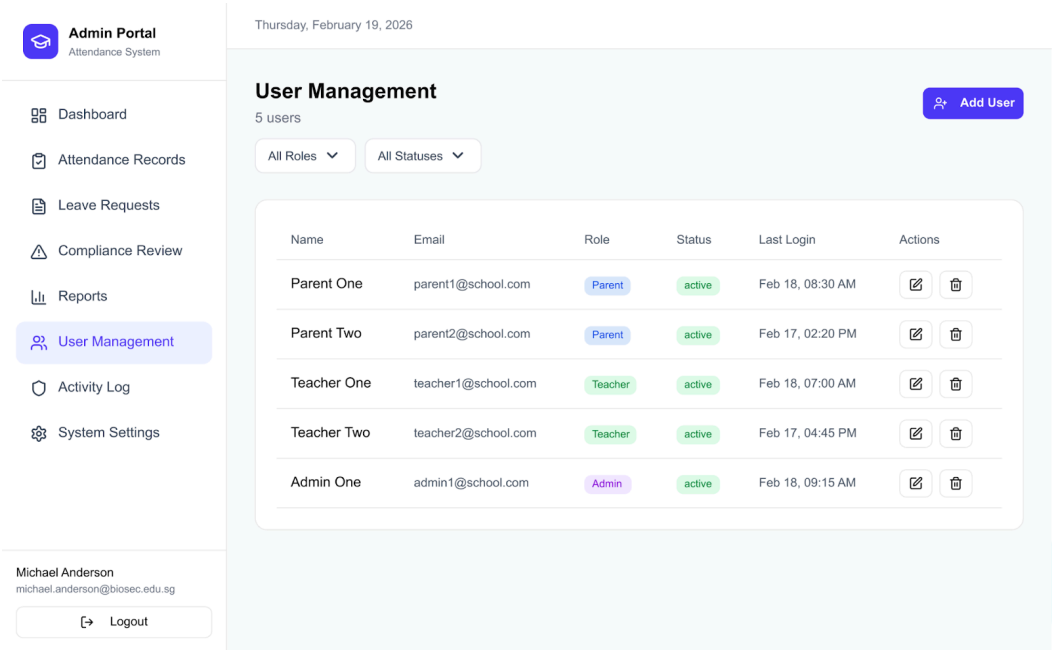


Figure 2.23: Administrator User Management Screen — BioSec School Attendance (UR-WEB-06, 07/02/2026)

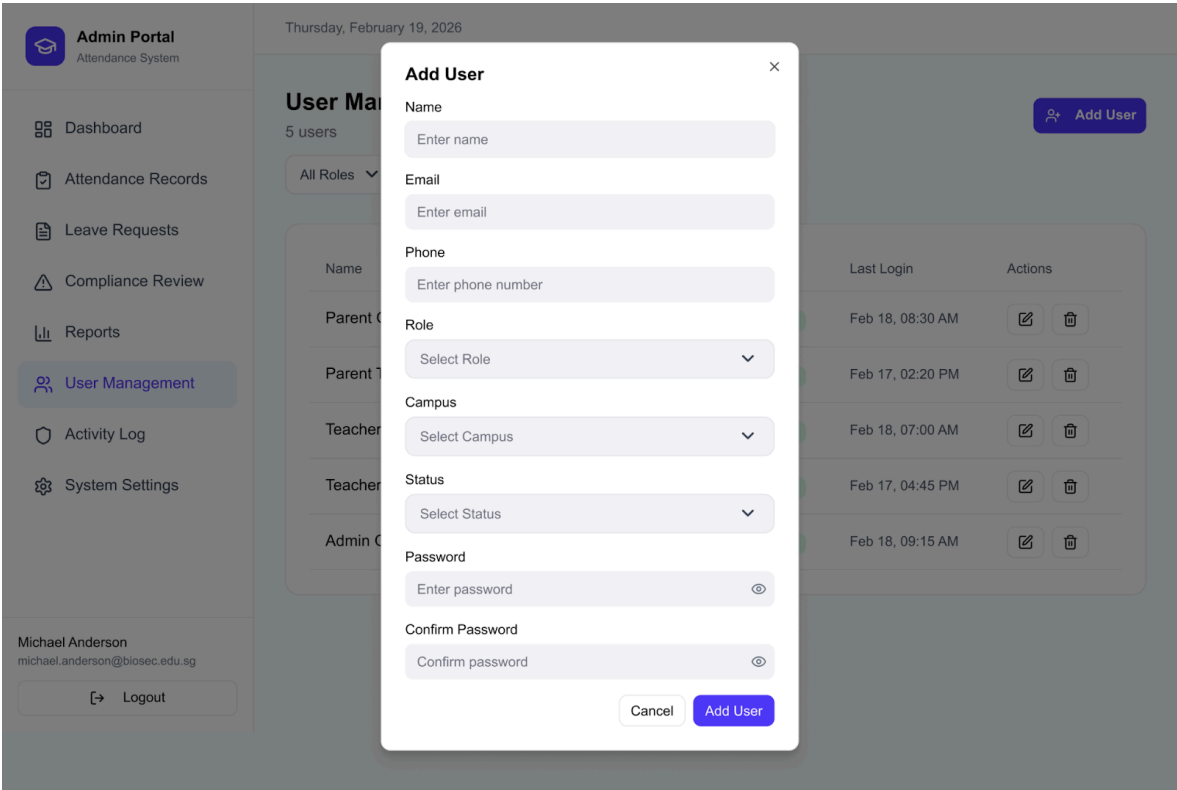


Figure 2.24: Administrator Create User Screen — BioSec School Attendance (UR-WEB-06, 07/02/2026)

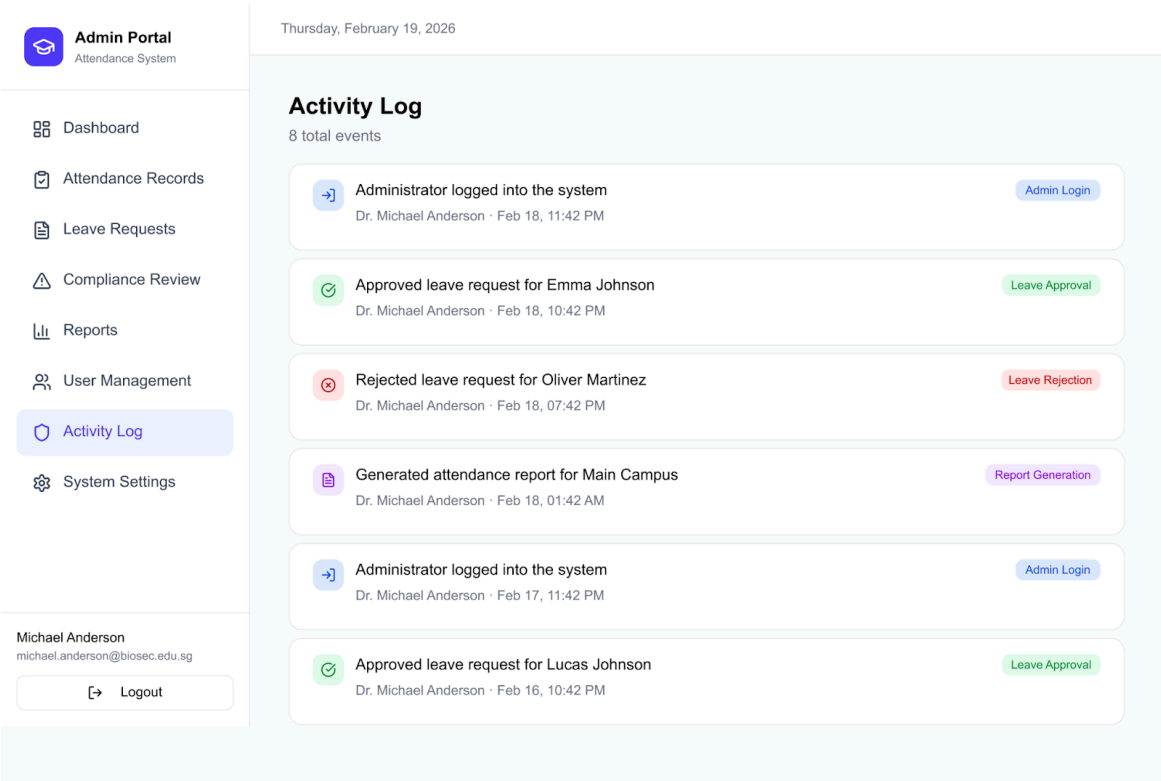


Figure 2.25: Administrator Activity Log Screen — BioSec School Attendance (UR-WEB-08, 07/02/2026)



 **Admin Portal**
Attendance System

Dashboard

Attendance Records

Leave Requests

Compliance Review

Reports

User Management

Activity Log

System Settings

Michael Anderson
michael.anderson@biosec.edu.sg

Logout

Thursday, February 19, 2026

System Settings

Attendance Settings

Compliance Threshold (%)
80
Students below this threshold will be flagged for compliance review

Warning Threshold (%)
85
Students below this threshold will receive attendance warnings

Notification Settings

Admin Notification Email
admin@school.com
System notifications will be sent to this email

Save Settings

Figure 2.26: Administrator System Setting Screen — BioSec School Attendance (UR-WEB-07, 07/02/2026)

Figure 2 – 12



Figure 3 Use Case Diagram

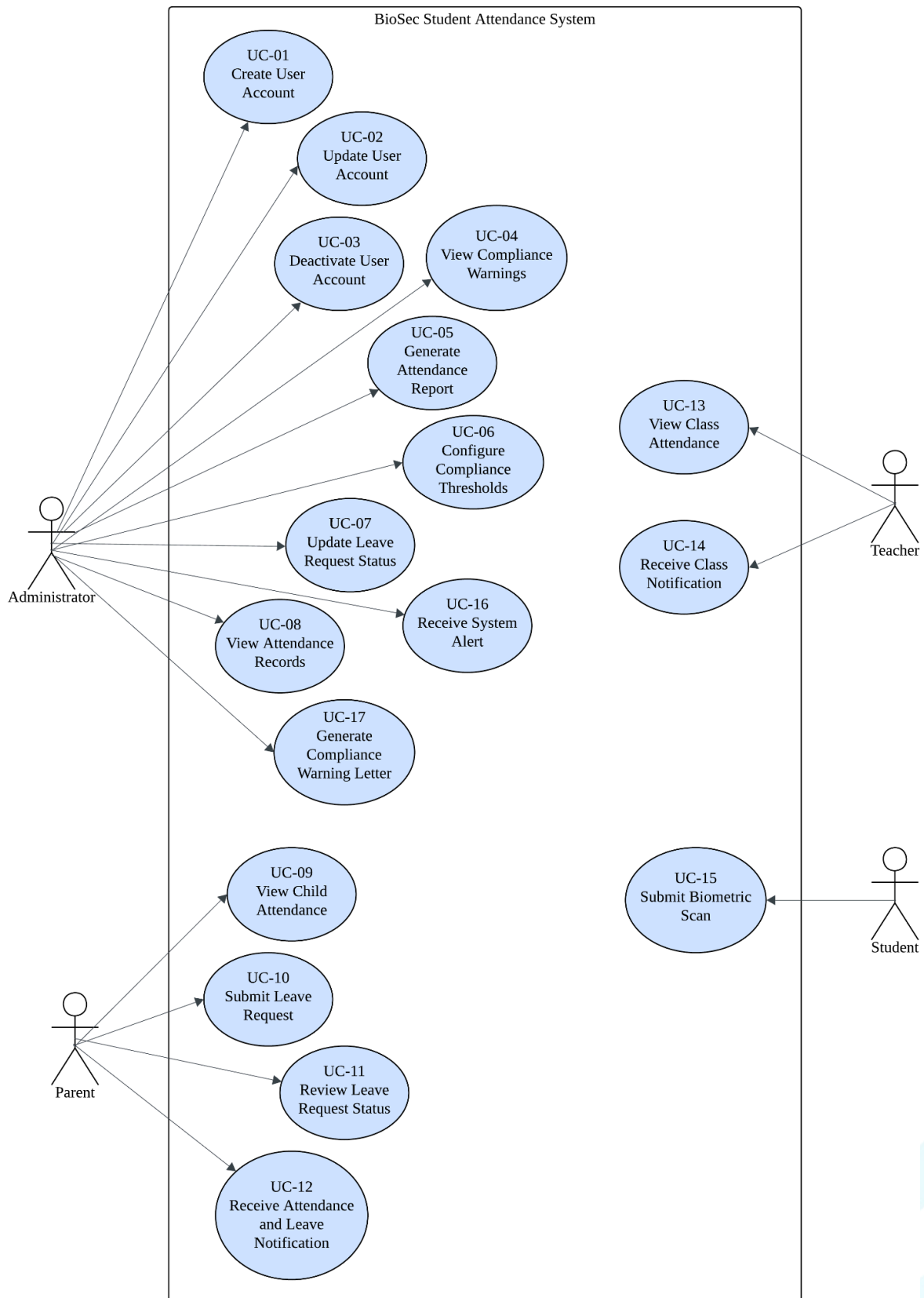


Figure 3: Use Case Diagram



Figure 4 ERD Diagram

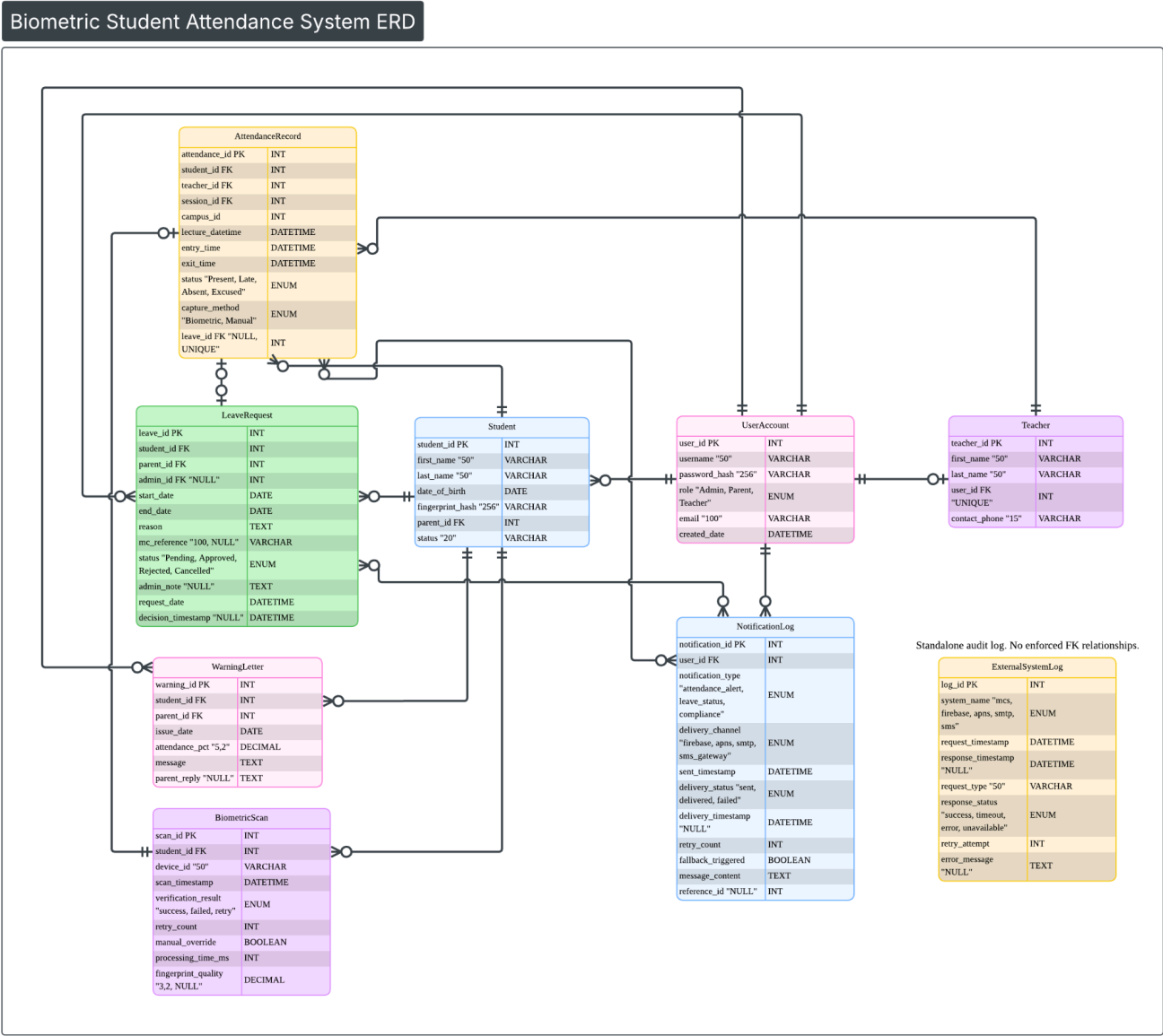


Figure 4: ERD Diagram



Appendix A

Glossary

Term	Definition
AES-256	Advanced Encryption Standard with 256-bit key length for biometric data encryption at rest.
APNS	Apple Push Notification Service for iOS push notifications.
API	Application Programming Interface for system integrations.
AR	Analysing Report (NS-AR-2026-001) by BioSec.
Azure SQL	Microsoft Azure cloud-based relational database service.
Biometric Template	Digital representation of fingerprint stored for authentication comparison.
BOR	Business Operations Requirement from the PS.
CAPEX	Capital Expenditure; one-time project investment costs.
DFD	Data Flow Diagram showing system data flows and external entities.
FAR	False Acceptance Rate; target <0.01% per TC-02 revision.
Firebase	Google Firebase Cloud Messaging for Android push notifications.
FR	Functional Requirement specifying system behaviour.
FRR	False Rejection Rate; target <1%.
IEEE 830	IEEE Std 830-1998 for SRS structure (basis of this template).
IRTM	Inceptive Requirements Traceability Matrix; PS to SRS chain of custody.
ISO/IEC 19794-2	International standard for fingerprint minutiae data interchange.
MC	Medical Certificate for illness/injury documentation.
MCS	Medical Certificate System; external integration for MC verification.
NFR	Non-Functional Requirement specifying quality or constraints.
OAuth 2.0	Authorization framework for secure delegated API access.
PDPA	Personal Data Protection Act 2012 (Singapore).
PS	Project Specification (BS-PS-2026-001) by BioSec.
RPO	Recovery Point Objective; maximum acceptable data loss duration.
RTO	Recovery Time Objective; maximum acceptable downtime duration.
SLA	Service Level Agreement between parties.



SMTP	Simple Mail Transfer Protocol for email notifications.
SRS	Software Requirements Specification (this document, NS-SRS-2026-001).
TLS	Transport Layer Security for encrypted communications.
TR	Technical Requirement from the PS.

Table A.1 – Glossary of Terms



Appendix B

Issues List

This appendix consolidates all unresolved TBD (to be determined) references that remain in this SRS. These correspond to the open issues defined under Change Mitigation Strategy CM#2 (Section 8.2). Full details including descriptions, fallback actions, rationale, and kill dates are provided in Table 2 of Section 8.2.

Issue ID	Issue Title	Status	Kill Date	SRS Reference(s)
CM-OI-01	MCS Integration Security	Open	20 Mar 2026	Section 2.7; Section 3.3; FR-SW-01; FR-SW-02; NFR-SEC-05
CM-OI-02	Disaster Recovery RPO/RTO	Open	20 Mar 2026	Section 2.7; NFR-DB-01; NFR-DOC-03
CM-OI-03	Biometric Template Format	Open	15 Mar 2026	Section 2.5; Section 3.2; NFR-DB-03

Table B.1 – Open Issues Summary



Appendix C

Data Dictionary

UserAccount

Field name	Data type	Size	Allow null?	Description	Example
user_id	INT	11	No	PK: Unique account ID	1001
username	VARCHAR	50	No	Login username	admin1
password_hash	VARCHAR	256	No	Encrypted password	(SHA256 hash)
role	ENUM	–	No	'Admin', 'Parent', 'Teacher'	'Parent'
email	VARCHAR	100	No	Registered email for notifications	parent@school.sg
created_date	DATETIME	–	No	Account creation timestamp	2026-01-01 10:00:00

Table C.1 – UserAccount Data Dictionary

Student

Field name	Data type	Size	Allow null?	Description	Example
student_id	INT	11	No	PK: Unique student ID	2026001
first_name	VARCHAR	50	No	Given name	Ah
last_name	VARCHAR	50	No	Family name	Mei
date_of_birth	DATE	–	No	Birth date	2010-05-15
fingerprint_hash	VARCHAR	256	No	Biometric hash for attendance matching	(fingerprint hash)
parent_id	INT	11	No	FK to UserAccount (parent account)	2001
status	VARCHAR	20	Yes	Active, Inactive, Suspended	Active

Table C.2 – Student Data Dictionary



Teacher

Field name	Data type	Size	Allow null?	Description	Example
teacher_id	INT	11	No	PK: Unique teacher ID	T001
first_name	VARCHAR	50	No	Given name	Mr. Tan
last_name	VARCHAR	50	No	Family name	Wei
user_id	INT	11	No	FK to UserAccount	3001
contact_phone	VARCHAR	15	Yes	Phone for notifications	+6581234567

Table C.3 – Teacher Data Dictionary

AttendanceRecord

Field name	Data type	Size	Allow null?	Description	Example
attendance_id	INT	11	No	PK: Unique record ID	500001
student_id	INT	11	No	FK to Student	2026001
teacher_id	INT	11	No	FK to Teacher (who took attendance)	T001
session_id	INT	11	No	FK to scheduled academic session	001
campus_id	VARCHAR	20	No	Campus identifier	Campus-01
lecture_datetime	DATETIME	–	No	Scheduled lecture start	2026-02-03 09:00:00
entry_time	DATETIME	–	Yes	Biometric entry scan time	2026-02-03 09:01:30
exit_time	DATETIME	–	Yes	Biometric exit scan time	2026-02-03 10:25:45
status	ENUM	–	No	Present, Late, Absent, Excused	Present



capture_method	ENUM	–	No	Types of capture methods (“Biometric” or “Manual”)	Biometric
leave_id	INT	11	Yes	FK to LeaveRequest (if Excused)	7001

Table C.4 – AttendanceRecord Data Dictionary

LeaveRequest

Field name	Data type	Size	Allow null?	Description	Example
leave_id	INT	11	No	PK: Unique leave request ID	7001
student_id	INT	11	No	FK to Student	2026001
parent_id	INT	11	No	FK to UserAccount (requester)	2001
admin_id	INT	11	Yes	FK to UserAccount (admin who approved/rejected)	1001
start_date	DATE	–	No	Leave start date	2026-02-05
end_date	DATE	–	No	Leave end date	2026-02-07
reason	TEXT	–	No	Brief reason (e.g., illness)	Sick leave
mc_reference	VARCHAR	100	Yes	MCS system reference for medical cert	MCS-2026-001
status	ENUM	–	No	Pending, Approved, Rejected, Cancelled	Approved
admin_note	TEXT	–	Yes	Admin response/comments	Approved with MC
request_date	DATETIME	–	No	Timestamp of request	2026-02-04 14:30:00
decision_timestamp	DATETIME	-	Yes	When admin approved or rejected	2026-02-04 17:15:00

Table C.5 – LeaveRequest Data Dictionary

WarningLetter



Field name	Data type	Size	Allow null?	Description	Example
warning_id	INT	11	No	PK: Unique warning ID	8001
student_id	INT	11	No	FK to Student	2026001
parent_id	INT	11	No	FK to UserAccount (recipient)	2001
issue_date	DATE	–	No	Date warning issued	2026-02-10
attendance_pct	DECIMAL	5,2	No	Attendance percentage triggering warning	75.50
threshold_value	DECIMAL	5,2	No	Compliance threshold percentage at time of warning	80.00
message	TEXT	–	No	Warning content (attendance <80%)	Improve attendance
parent_reply	TEXT	–	Yes	Parent response	Will monitor

Table C.6 – WarningLetter Data Dictionary

NotificationLog

Field name	Data type	Size	Allow null?	Description	Example
notification_id	INT	11	No	PK: Unique notification ID	5001
user_id	INT	11	No	FK to UserAccount (recipient)	2001
notification_type	ENUM	–	No	'attendance_alert', 'leave_status', 'compliance'	attendance_alert
delivery_channel	ENUM	–	No	'firebase', 'apns', 'smtp',	firebase



				'sms_gateway'	
sent_timestamp	DATETIME	–	No	When notification sent	2026-02-10 09:15:00
delivery_status	ENUM	–	No	'sent', 'delivered', 'failed'	delivered
delivery_timestamp	DATETIME	–	Yes	When delivery confirmed	2026-02-10 09:15:23
retry_count	INT	2	No	Number of retry attempts	1
fallback_triggered	BOOLEAN	–	No	SMS fallback activated (30min)	0
message_content	TEXT	–	No	Notification message body	Your child was absent today
reference_id	INT	11	Yes	FK to related record	3001

Table C.7 – Notification Data Dictionary

BiometricScan

Field name	Data type	Size	Allow null?	Description	Example
scan_id	INT	11	No	PK: Unique scan ID	600001
student_id	INT	11	No	FK to Student	2026001
device_id	VARCHAR	50	No	Biometric device identifier	BIO-DEVICE-01



scan_timestamp	DATETIME	–	No	When scan occurred	2026-02-10 09:01:15
verification_result	ENUM	–	No	'success', 'failed', 'retry'	success
retry_count	INT	2	No	Number of retry attempts	0
manual_override	BOOLEAN	–	No	Manual attendance authorized	0
processing_time_ms	INT	6	No	Scan processing time in ms	1850
fingerprint_quality	DECIMAL	3,2	Yes	Quality score (0-1)	0.92

Table C.8 – BiometricScan Data Dictionary

ExternalSystemLog

Field name	Data type	Size	Allow null?	Description	Example
log_id	INT	11	No	PK: Unique log entry ID	11001
system_name	ENUM	–	No	'mcs', 'firebase', 'apns', 'smtp', 'sms'	mcs
request_timestamp	DATETIME	–	No	When request sent	2026-02-10 09:20:00
response_timestamp	DATETIME	–	Yes	When response received	2026-02-10 09:20:04
request_type	VARCHAR	50	No	API endpoint/operation	validate_mc
response_status	ENUM	–	No	'success', 'timeout', 'error', 'unavailable'	success



retry_attempt	INT	2	No	Retry attempt number	1
error_message	TEXT	–	Yes	Error details if failed	Connection timeout

Table C.9 – ExternalSystemLog Data Dictionary

Appendix D

Inceptive Requirements Traceability Matrix (iRTM)

Document type: Inceptive Requirements Traceability Matrix (iRTM)

Project ID: NS-PROJ-2026-001

Project Name: Biometric Student Attendance System - Nevon Solutions (NS)

Document date: 3 March 2026

Assumption(s): Platform-level constraints (BOR-0-xx, TR-0-xx) are precursors applicable to the entire system. All requirements from PS (BS-PS-2026-001) have undergone regulated-revision via the Analysing method before inclusion in this SRS.

Constraint(s): All biometric data handling must comply with PDPA 2012.

Regulation(s): Singapore Personal Data Protection Act (PDPA) 2012, as amended.

Ser	Project Specification					Software Requirement Specification				
	Category	ID	Desc	Stakeholder	Precedence	Category	ID	Desc	Owner	Ranking
1	Biz Ops	BOR-0-01	System shall enforce role-based access control for all user classes across all campuses	BioSec PM	Essential	System Features	FR-RBAC-01	RBAC for three user classes enforced at application and API layers.	Dev Team	High
							FR-RBAC-02	Admin creates user accounts with unique ID, role, and linked entity.		
							FR-RBAC-03	Admin updates role assignment, resets credentials, unlocks accounts. Changes logged.		
							FR-RBAC-04	Admin deactivates accounts. Denied login immediately, kept for audit.		
2	Biz Ops	BOR-0-02	Each user class shall access only functions and data required for their role	BioSec PM	Essential	System Features	FR-RBAC-01	Role-scoped access consistent across all five campuses. Enforced at application and API layers.	Dev Team	High
3	Biz Ops	BOR-0-03	Only biometric scan results are valid attendance records; manual override after 3 failed scans; scanner failure handling	BioSec Ops	Essential	External Interface Requirements	FR-HW-01	The system receives scan data from fingerprint devices at five campuses, each identified by unique device_id.	Dev Team	High
						Business Rules	FR-HW-04	After 3 consecutive failed scans, supervised manual override initiated with flagged record for admin review.		

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							FR-HW-05	Scanner failure does not auto-mark absent. Failure logged with device_id and timestamp.		
							BR-01	Only biometric scan results are valid. Manual recording not permitted except via supervised override.		
							FR-NOTIF-08	Admin notified of scanner device failure		
4	Biz Ops	BOR-1-MOB-01	Parents and teachers must authenticate before accessing attendance features	BioSec PM	Essential	System Features	FR-AUTH-01	Credentials validated before enabling access to attendance functionality.	Dev Team	High
							FR-AUTH-02	Session token generated on success, terminated after configurable inactivity.		
5	Biz Ops	BOR-1-MOB-02a BOR-1-MOB-02b BOR-1-MOB-02c	Parents view own child attendance; teachers view assigned classes; mobile sync every 5 min	BioSec PM	Essential	System Features Communications Interface	FR-ATT-01	Parents view attendance for registered child only, enforced at application and API layers.	Dev Team	High
							FR-ATT-02	Teachers view attendance for assigned classes only, enforced at application and API layers.		
							FR-ATT-03	AttendanceRecord created per scan storing student ID, session ID, campus, timestamp, capture method.		
							FR-ATT-04	No deletion of AttendanceRecord entries. Deletion attempts rejected and recorded.		
							FR-COM-01	Mobile sync every 5 min when connected. Max data staleness 30 min.		
6	Biz Ops	BOR-1-MOB-03a BOR-1-MOB-03b	Parents submit leave requests with MCS medical certificate validation	BioSec PM	Essential	System Features	FR-LEAVE-01	Authenticated parents submit leave requests for registered child.	Dev Team	High
							FR-LEAVE-02	MC submitted to MCS for validation before recording. Leave not recorded if MCS fails.		
							FR-LEAVE-03	All submissions logged with timestamp, user ID, student ID, leave reason, MCS outcome.		

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							FR-LEAVE-04	Parents view status of all leave requests including MCS outcome and admin decision.		
7	Biz Ops	BOR-1-MOB-04	System shall send push notifications for absence events and leave status updates within 15 minutes	BioSec PM	Essential	System Features	FR-NOTIF-01	Parents notified via push within 15 min of missed attendance session.	Dev Team	High
							FR-NOTIF-02	Parents notified via push of leave request status changes (approved/rejected).		
							FR-NOTIF-03	Notification record created per dispatch with recipient, type, content, timestamp, channel, status.		
							FR-NOTIF-04	Parents view notification history ordered by most recent.		
							FR-NOTIF-05	Delivery status updated when confirmed by Firebase, APNS, SMS gateway, or fallback activated.		
							FR-NOTIF-06	Notification records retained at least 12 months before archive or purge.		
							FR-NOTIF-09	Teachers notified via push of attendance events for assigned classes.		
							NFR-PERF-04	Push notifications within 15 minutes		
8	Biz Ops	BOR-1-WEB-01a BOR-1-WEB-01b	Administrator web portal for attendance oversight and compliance monitoring across five campuses	BioSec Admin	Essential	System Features	FR-OVS-01	Web portal displays attendance for all students across five campuses, filterable by campus, class, student, date.	Dev Team	High
		FR-OVS-02					Admin configures compliance threshold percentage. Changes immediate, logged with admin ID and timestamp.			
		FR-OVS-03					System flags students whose attendance falls below configured threshold.			
		FR-OVS-04					Admin delivers compliance warning letter to parent email of flagged students.			
		FR-OVS-05					WarningLetter record generated per letter			

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							FR-OVS-06	Attendance reports saved 12 months. No deletion. System archives on regular basis.		
9	Biz Ops	BOR-1-WEB-02a BOR-1-WEB-02b	Administrators review, approve/reject leave requests with MCS validation	BioSec Admin	Essential	System Features	FR-LVR-01	Admin retrieves and views all pending leave requests with MCS validation outcome.	Dev Team	High
							FR-LVR-02	Approval blocked if MC not validated by MCS.		
							FR-LVR-03	Attendance record updated with approved leave status upon admin approval.		
							FR-LVR-04	LeaveRequest status updated to Approved/Rejected with admin ID, timestamp, comments.		
							FR-LVR-05	No deletion of LeaveRequest records. All kept for audit.		
							FR-NOTIF-07	Admin notified of new pending leave request		
10	Biz Ops	BOR-1-ATT-01	Attendance capture only during scheduled sessions; scanner failure shall not mark students absent	BioSec Ops	Essential	Business Rules Nonfunctional Requirements	BR-01	Only biometric scan results are valid attendance records. Manual override per FR-HW-04 only.	Dev Team	High
							BR-02	Attendance capture only during scheduled academic sessions. Scans outside session windows rejected.		
							NFR-SAF-02	Scanner failure shall not auto-mark absent. Prevents incorrect attendance penalties.		
11	IT Tech	TR-0-01	System shall support Android (min 10) and iOS (min 14) mobile apps with push notification via Firebase and APNS	BioSec IT	Essential	Software Interfaces Software Quality Attributes	FR-SW-03	Push notifications via FCM for Android and APNS for iOS, covering attendance and leave alerts.	Dev Team	High
							FR-COM-02	Push via Firebase and APNS. SMS fallback if undelivered after 30 min.		
							QA-PORT-01	Native Android 10+ and iOS 14+ apps. Backend deployable on Azure without platform-specific changes.		
12	IT Tech	TR-0-02	System shall be hosted on Microsoft Azure Singapore region with HTTPS and TLS 1.2+	BioSec IT	Essential	Nonfunctional Requirements Software Quality	NFR-PERF-05	Support 1,500 concurrent sessions. Graceful degradation to 2,000 at max 4s response. Auto-scale at 70% CPU.	Dev Team	High

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						Attributes	QA-PORT-01	Backend deployable on Microsoft Azure without platform-specific code changes.		
13	IT Tech	TR-1-BE-01b	System shall meet performance thresholds for API response, scan processing, and report generation	BioSec IT	Essential	Nonfunctional Requirements	NFR-PERF-01	Web portal pages load within 3s under normal conditions.	Dev Team	High
							NFR-PERF-02	95% of API responses within 500ms under normal load.		
							NFR-PERF-03	Reports within 10s for single class, 60s for institution-wide.		
							NFR-PERF-05	1,500 concurrent user sessions without compromising response time or scan throughput.		
14	IT Tech	TR-1-BE-02	MCS integration via HTTPS REST API with OAuth 2.0; 5s timeout, 3 retries with exponential backoff	BioSec IT	Essential	Software Interfaces Nonfunctional Requirements	FR-SW-01	MC validated via MCS REST API using HTTPS and OAuth 2.0 before recording leave.	Dev Team	High
							FR-SW-02	5s timeout, 3 retries with exponential backoff. Failures logged. Admin notified on exhaustion.		
							FR-COM-03	MCS API communication per interface contract in FR-SW-01 and FR-SW-02.		
							NFR-SEC-05	All MCS API via HTTPS and OAuth 2.0. No alternative auth mechanism permitted.		
15	IT Tech	TR-1-BE-03	All data encrypted AES-256 at rest and TLS 1.2+ in transit; immutable audit logs; PDPA compliance	BioSec IT	Essential	Nonfunctional Requirements Business Rules	NFR-SEC-01	Biometric data encrypted AES-256 at rest, TLS 1.2+ in transit. No plaintext at any point.	Dev Team	High
							NFR-SEC-02	All PII processed and stored in accordance with PDPA retention policy.		
							NFR-SEC-06	Audit logs retained 12 months. Protected from modification or deletion.		
							NFR-SAF-01	Prohibit overwriting/deleting attendance records		
							NFR-SAF-03	All biometric data encrypted AES-256 at rest and TLS 1.2+ in transit.		

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							NFR-SAF-04	Audit logs immutable once written. No user may modify or delete entries.		
							BR-08	Biometric data for attendance only. Not shared with third parties without legal permission.		
							BR-09	Audit logs are official record. No user role may modify or delete entries.		
							NFR-LEGAL-01	PDPA compliance		
16	IT Tech	TR-1-BE-04	Biometric scan processing within 2 seconds; 50 students per device per 10 minutes throughput	BioSec IT	Essential	External Interface Requirements Software Quality Attributes	FR-HW-02	Scan processing ≤2s per attempt. 50 students/device/10 min during peak.	Dev Team	High
							FR-HW-03	200 concurrent scans across all campus devices without exceeding 2s or degrading API response.		
							QA-EFFIC-01	Scans in 2s, API in 500ms for 95% of calls. No redundant DB queries per transaction.		
17	IT Tech	TR-1-BE-05a TR-1-BE-05b TR-1-BE-05c	Microsoft SQL Server on Azure SQL Database; AES-256 biometric encryption; daily backup with monthly restoration verification	BioSec IT	Essential	Software Interfaces Other Requirements	FR-SW-05	Azure SQL Database with AES-256 for biometric data. Daily backups. Monthly restoration verification.	Dev Team	High
							NFR-DB-01	Microsoft SQL Server on Azure SQL Database. Daily automated backups, monthly restoration verification.		
							NFR-DB-02	Referential integrity enforced. Duplicates detected and rejected at application and database layers.		
							NFR-DB-03	All biometric templates encrypted AES-256. No plaintext biometric data written to the database.		
18	IT Tech	TR-1-CL-01a TR-1-CL-01b	Mobile apps distributed via Google Play Store and Apple App Store	BioSec IT	Essential	Software Quality Attributes Nonfunctional Requirements	QA-PORT-01	Native Android 10+ and iOS 14+ apps. Backend on Azure without platform-specific changes.	Dev Team	High
							NFR-SEC-08	Apps code-signed with BioSec institutional certificates before store distribution.		
19	IT Tech	TR-1-CL-01c	Mobile apps shall sync attendance data every 5 minutes with max 30 min staleness	BioSec IT	Essential	Communications Interfaces	FR-COM-01	Sync every 5 min when connected. Max data staleness 30 min.	Dev Team	High

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20	IT Tech	TR-1-CL-01d	Mobile apps must be code-signed with BioSec institutional certificates	BioSec IT	Essential	Nonfunctional Requirements	NFR-SEC-08	Code-signed with BioSec institutional certificates before Google Play and App Store distribution.	Dev Team	High
21	IT Tech	TR-1-CL-01e	Support for new OS versions within 30 days of public release during warranty	BioSec IT	Desirable	Other Requirements	NFR-MAINT-01	New OS version support within 30 days during warranty. All updates backward compatible.	Dev Team	Medium
22	IT Tech	TR-1-CL-01f	Offline viewing and leave drafting for up to 48 hours; 50MB local storage limit; AES-256 local encryption	BioSec IT	Essential	Nonfunctional Requirements	NFR-SEC-09	Local cache encrypted AES-256. Auto-purge after 48 hours. 50MB local storage limit.	Dev Team	High
23	IT Tech	TR-1-CL-02	Web portal access restricted to campus networks or secure VPN	BioSec IT	Essential	Nonfunctional Requirements	NFR-SEC-07	Admin portal restricted to campus networks or VPN. No direct internet access.	Dev Team	High
24	IT Tech	TR-1-CPR-01	Two-factor authentication for administrators; session management; account lockout policy	BioSec IT	Essential	Nonfunctional Requirements	NFR-SEC-03	2FA enforced for all admin accounts accessing the web portal.	Dev Team	High
							NFR-SEC-04	Account locked after configurable failed login attempts. Configurable lockout duration. Events logged.		
							NFR-SEC-06	Audit logs retained 12 months. Protected from modification or deletion.		
25	IT Tech	TR-1-CPR-03	Referential integrity; duplicate student ID detection at application and database layers	BioSec IT	Essential	Other Requirements	NFR-DB-02	Referential integrity enforced. Duplicates rejected at application and database layers.	Dev Team	High
							NFR-MIG-01	Legacy data migration from Excel/CSV. All current year data migrated and validated before go-live.		
							NFR-MIG-02	Duplicate student IDs flagged during migration. Duplicates not imported, reported to BioSec.		
26	IT Tech	TR-1-CPR-04	Three-tier severity maintenance model during 12-month warranty period	BioSec IT	Essential	Software Quality Attributes	QA-MAINT-01	Critical: 4h response/24h resolution. High: 8h/48h. Medium: 24h/5-day. 12-month warranty.	Dev Team	High
						Other Requirements	NFR-MAINT-01	New OS version support within 30 days during warranty. Backward compatible updates.		

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27	IT Tech	TR-1-DOC-01	Delivery of user manuals, API docs, database docs, source code docs, and architecture diagrams	BioSec IT	Essential	Other Requirements	NFR-DOC-01	Admin/Teacher/Parent manuals in PDF with screenshots and campus-specific procedures.	Dev Team	High	Go to Design Stage
		TR-1-DOC-02					NFR-DOC-02	API docs in Swagger/OpenAPI. DB schema with ERD and stored procedures.			
		TR-1-DOC-03					NFR-DOC-03	System architecture diagrams and deployment docs for expansion and disaster recovery.			
		TR-1-DOC-04					NFR-LEGAL-02	All source code, schemas, API docs, manuals, and architecture artifacts owned by BioSec.			

Table D.1 – Inceptive Requirements Traceability Matrix (PS to SRS)

Appendix E

Change Request Form

This appendix provides the standard Change Request Form template to be used for all change requests governed by the High-Risk Change Control Process defined in Section 8.3.



NEVON SOLUTIONS
Information Technology

CHANGE REQUEST FORM

Biometric Student Attendance System
NS-SRS-2026-001

CR Number	CR-_____	Date Submitted	DD/MM/YYYY
Project Name	Biometric Student Attendance System		
SRS Version	0.8		
Submitted By	Name	Role	e.g., Director of Operations
Contact	Email / Phone		
Priority	☐ High ☐ Medium ☐ Low		

Section 1: Change Request Details

Requirement ID	e.g., NFR-SEC-01
Current Wording	
Proposed Modification	
Reason for Change	
Expected Business Value	

Section 2: Impact Assessment (Nevon Solutions — within 5 working days)

Technical Feasibility	☐ Feasible ☐ Not Feasible ☐ Feasible with Conditions
Feasibility Comments	
Schedule Impact	Estimated delay (days/weeks)
Budget Impact	Estimated additional cost
Dependent Requirements	List affected requirement IDs
Risk Level	☐ High ☐ Medium ☐ Low
Assessed By	Name, Date



Change Request Form for Biometric Student Attendance System

Section 3: CCB Decision

Decision	☐ Approved ☐ Rejected ☐ Deferred
Conditions / Comments	
Schedule Extension	<i>Agreed extension (if any)</i>
Contract Amendment Required	☐ Yes ☐ No

	BioSec Representative	Nevon Solutions Representative
Name		
Signature		
Date		

Section 4: Implementation Tracking

SRS Section(s) Updated	<i>List section numbers</i>
iRTM Updated	☐ Yes ☐ No ☐ N/A
Implementation Date	<i>DD/MM/YYYY</i>
Verified By	<i>Name, Date</i>

*This form is governed by the High-Risk Change Control Process defined in Section 8.3 of NS-SRS-2026-001.
Change requests are valid only for requirements that already exist in the SRS.*